Approaching

for bass flute, bass clarinet, percussion, viola, cello and electronics

Justin Yang

The score for *Approaching* is in a computer-animated digital format. It is accessed and played like an online video game through a web browser.

Demo Recording with Score-Following Video: https://youtu.be/EMYu62G8KiM

Instrumentation

Bass flute, bass clarinet, percussion, viola, cello, electronics.

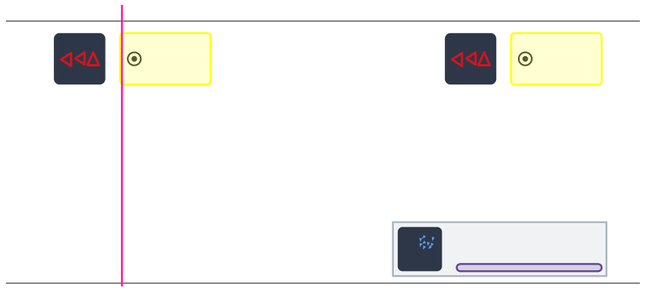
Percussion: crotales (bowed), used in the audible beats section; cymbals (bowed), recommended for the multiphonics section, but at the player's discretion. The rest of the instruments are chosen at the player's discretion.

Animated Conduction Tools

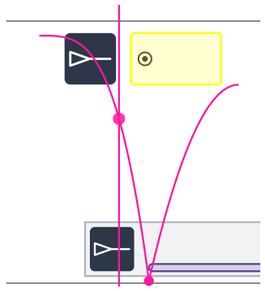
There are a number of animated conduction tools that are intuitive to follow and aid performers in realizing the score.

A **scrolling cursor** runs throughout the piece at a consistent speed and indicates when things are to be played. Notation is spaced proportionally in time — distance on the page equals duration.

Bouncing-ball conductors aid with rhythmic precision. Play at the precise moment of impact, using the ictus of the bouncing ball to inform the attack.

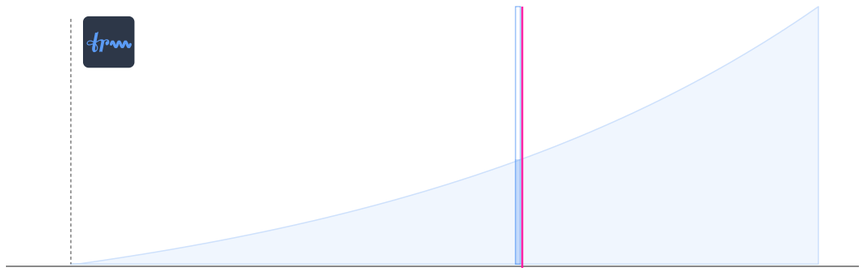


Animated scrolling cursor



Bouncing-ball conductor

Gradient Curves. This score uses graphic curves to describe continuous change over time. An animated curve follower appears alongside the cursor and gives the real-time position on the curve.



Language Music

This composition employs a version of Anthony Braxton's Language Music.

 **Short Attacks**

 **Trills**

 **Accented Long Tones**

 **Multiphonics**

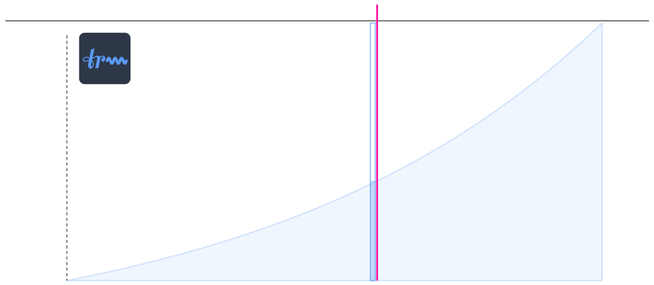
 **Audible Beats:** For this musical material, the live electronics will generate sine tones at specific pitches. The performer will generate audible beating against these sine tones using small glissandos around the unison.

 **Scattered Strikes:** Scattered strikes is an ensemble language type. The ensemble will produce a clustered impulse within a short time frame, like a handful of tossed pebbles hitting the ground.

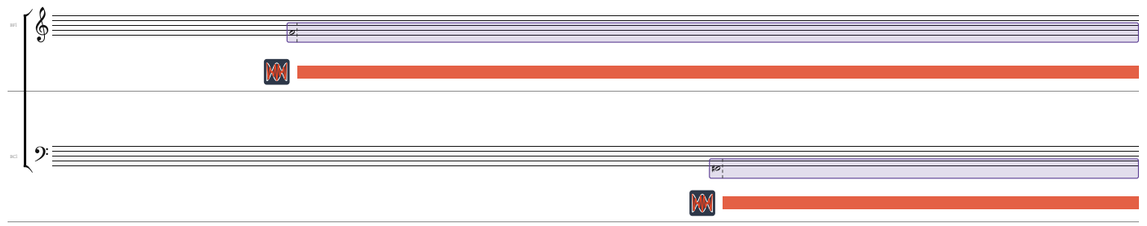
The use of language music is notated in different ways:



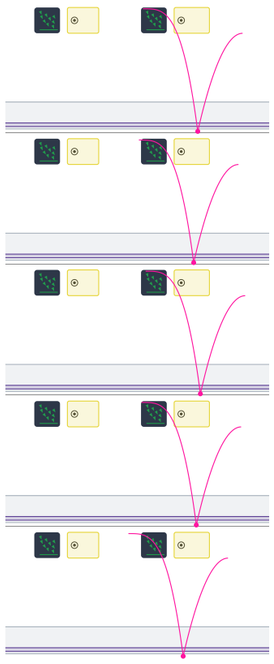
A language music badge next to a yellow box indicates that you would play this language type within the duration of the yellow box.



A language type with a gradient curve indicates that you would play that language type following the gradient of the curve. In the pictured example, the curve represents trill intensity. This could be a combination of speed and volume, and/or any other parameters a performer would use to develop a gradient of trill intensity.



A language music badge with an extended colored line indicates that you would play this language type freely throughout the duration of the line. Line thickness gives a general indication of density or intensity, to be interpreted freely. Each language type is assigned a color.

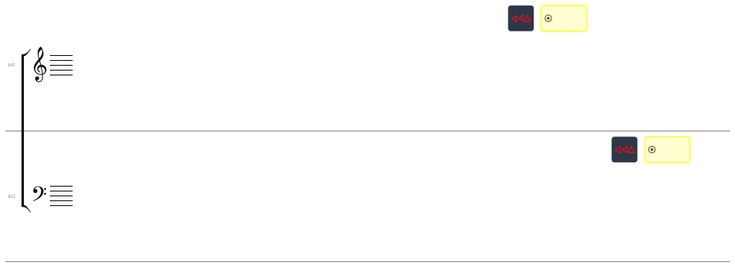


Scattered strikes are notated in two different ways. For some of the scattered strikes there is a specific rhythm notated for the strike using bouncing-ball conductors. Others have a time window within which a strike is to occur, indicated by a yellow duration box. In these cases, the ensemble would try to create a scattered strike within the time frame of the yellow box.

Live Electronics

Yellow boxes with a recording icon indicate a microphone opening. Your microphone will be live for the duration of the yellow box, and the live electronics will either record your input or use your input for live processing.

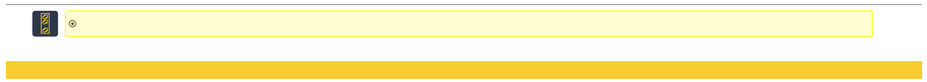
In section one of the piece, the short attacks section, there are a number of short attack windows. You would play one attack within each window. These attacks will be cropped by the system and used throughout the piece to generate live electronics.



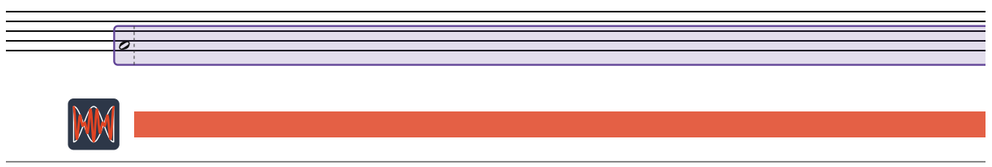
In section three of the piece, the trills and accented long tones section, a performer's accented long tone will activate resonant filters live, for the duration of the mic window.



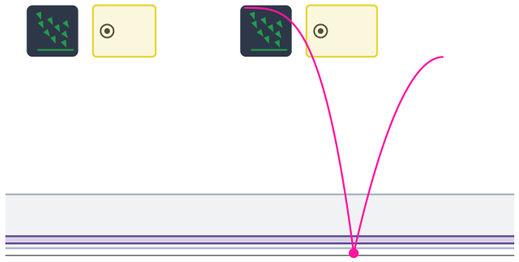
In section four, the multiphonics section, the mic openings will record your multiphonics for the duration of the yellow box into a buffer to be used for live processing throughout this section. These buffers will be used as samples that will be processed and played back, so everything you play for the duration of the box will be used for the live processing.



In section five of the piece, the audible beats section, the electronics will play sine tones at specific pitches. Each pitch is indicated in a purple box, and the electronics will play it for the duration of that box. Performers will then create audible beats against the sine tones. The percussion will use bowed crotales in this section.



In section six, the scattered strikes section, the mic openings will record the rhythm of the ensemble strike, and the recorded strike patterns will be used and transformed to produce the live electronics in this section.



Gray boxes are used in the full score to show the approximate time and type of live electronics that are playing.





Three-Body Problem (Section Two)

The three-body problem is a performance model that uses the three-body problem from physics as a framework to guide ensemble music interaction. Section two is based on this model.

In physics, the three-body problem asks how three bodies, three stars for example, move when each one pulls on the other two by gravity. Two bodies pulling on each other form perfectly predictable patterns, but with three bodies the predictability falls apart in an interesting way: deterministic chaos — a system that is completely determined but impossible to predict. When the three bodies are far enough apart they have very little effect on each other. As the bodies draw closer, their paths start to bend toward one another, and the motion of each body now depends on where the others are. In a close encounter, the bodies sweep past one another at high speed and pull hard on each other, and in that moment they trade energy. Finally, a body is ejected and it travels away on a long, slow loop until gravity brings it back into the cycle again.

The three-body problem musical model has four phases, each represented by its own badge:



Far Apart: Keep your own pace. Play as if solo and ignore the other activity.



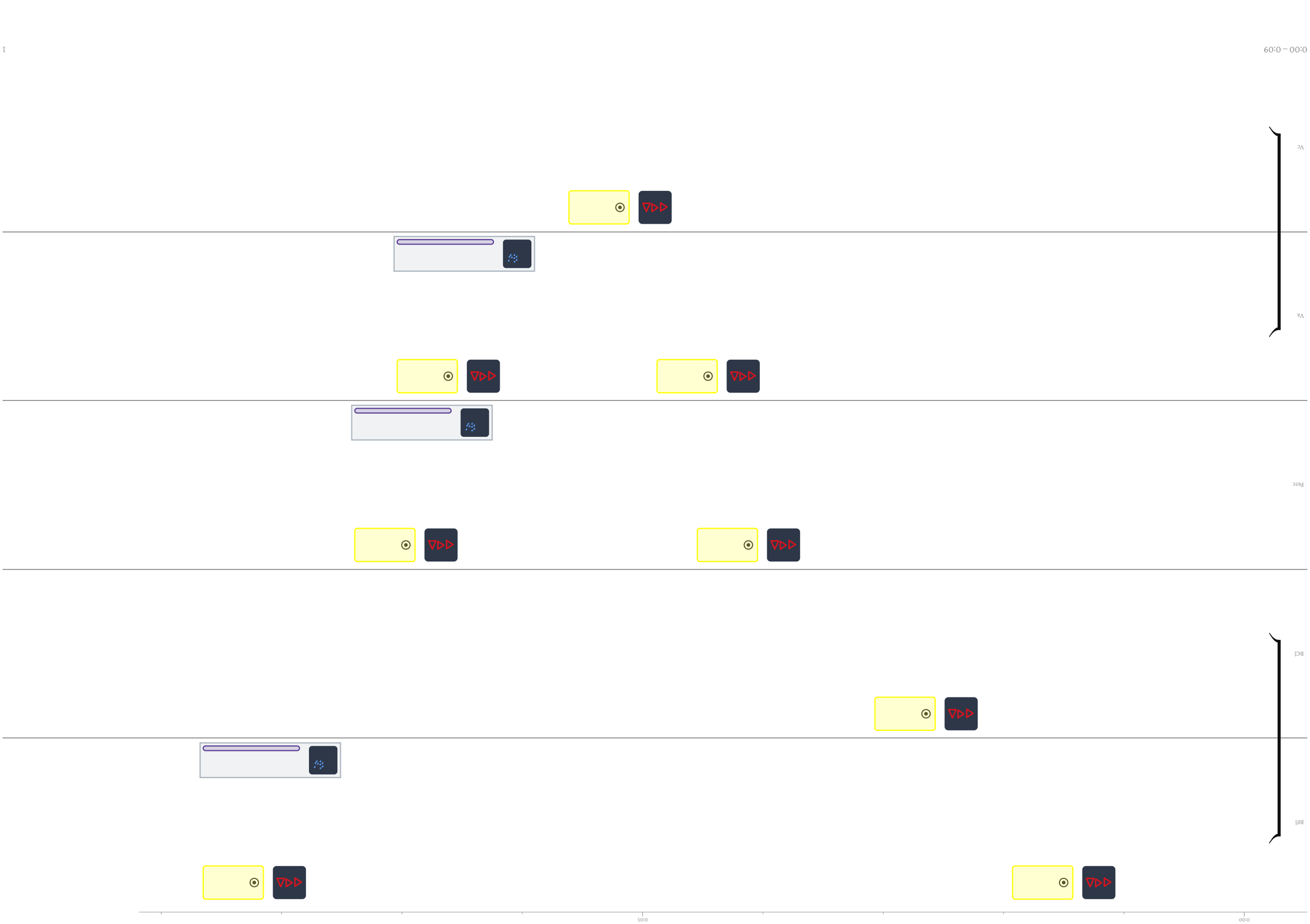
Approaching: Choose a player, or a particular sound to engage. When you hear it: wait a beat, then respond. Repeat — listen · hear · respond.



Close Pass: Choose a player, or several players in a cluster. Try to play just before or just after their onsets — as close to them as you can, without playing at the same time. You can respond or react to the energy in the cluster.

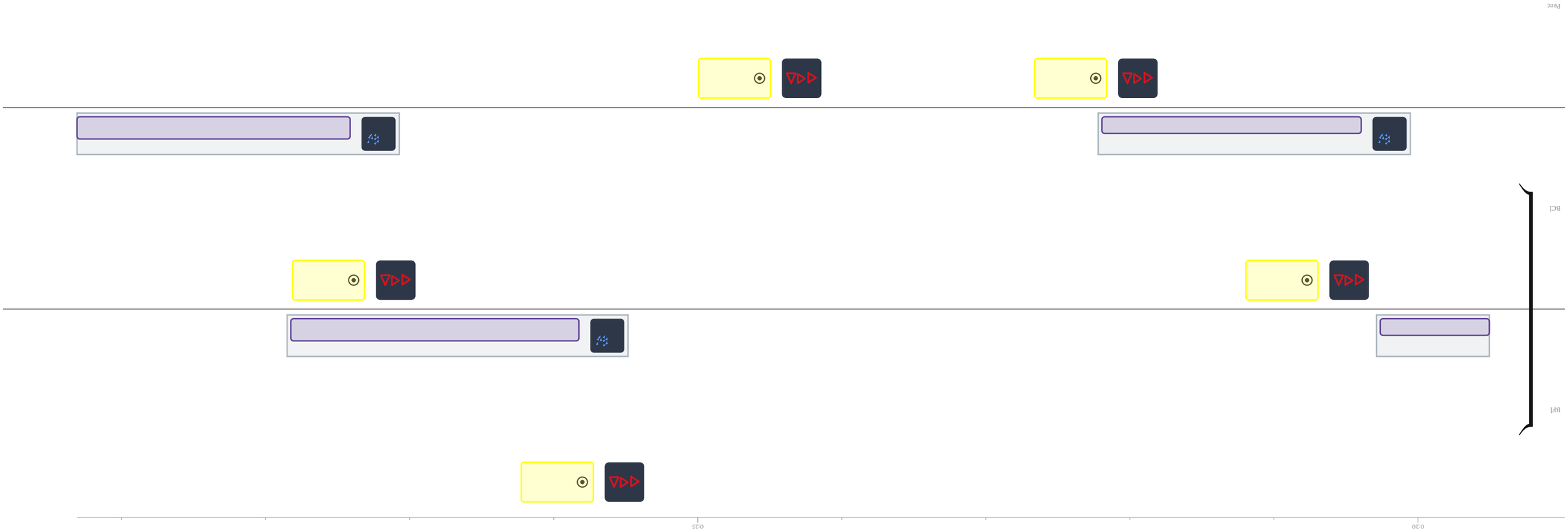


Break and Rejoin: Break, then, after a silence, rejoin. If the opportunity presents itself, try to rejoin the texture just after another cluster has broken up.


Petc
Va
Vc
0:00 – 0:09
1
BF1
BC1
0:00
0:05



0:09 – 0:20


Fetc
Va
Vc
0:20 – 0:29
3
BCl
BFl
0:20
0:25

Petc

Elec2

Va

Vc

Elec3

0:29 – 0:40

4

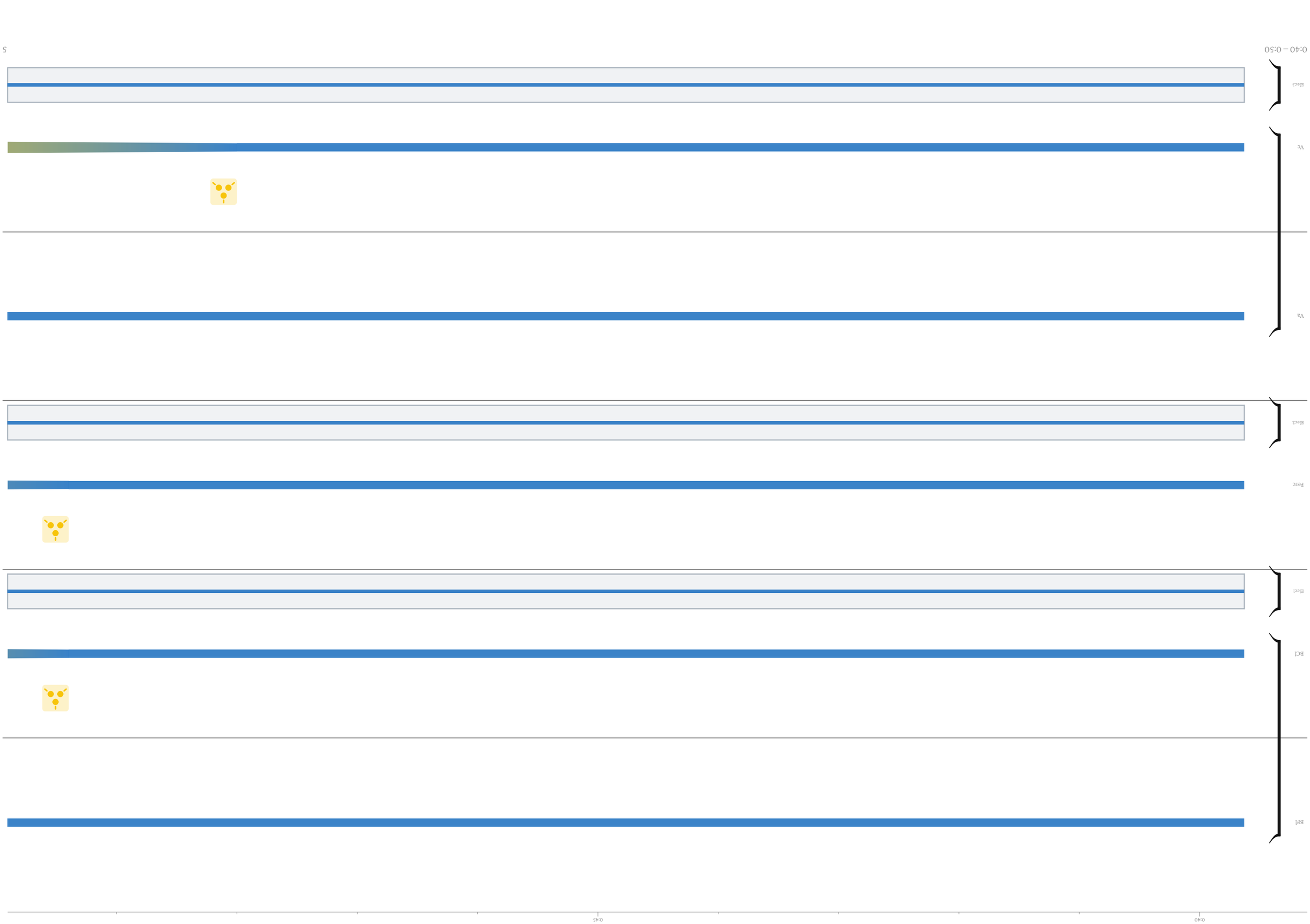
Elec1

BCl

BFl

0:30

0:35


Perc
Elec2
Va
Vc
Elec3
0:40 – 0:50
5
Elec1
BCl
BFl
0:40
0:45


0:50 – 1:00
Elec3
Vc
Va
Elec2
Petc
Elec1
BCl
BFl
0:50
0:55
1:00

1:05

1:10

BFl

BCl

Elec1

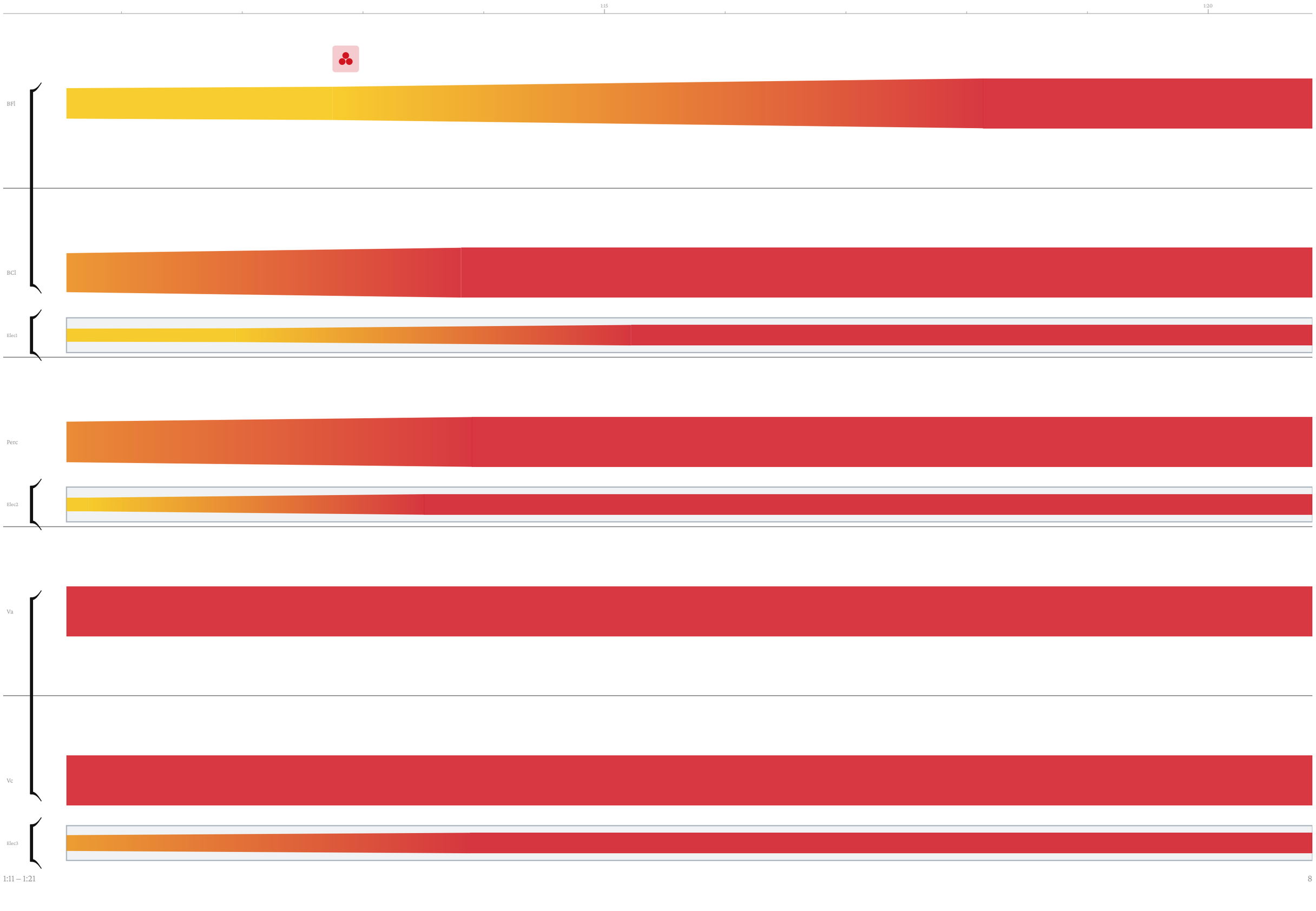
Perc

Elec2

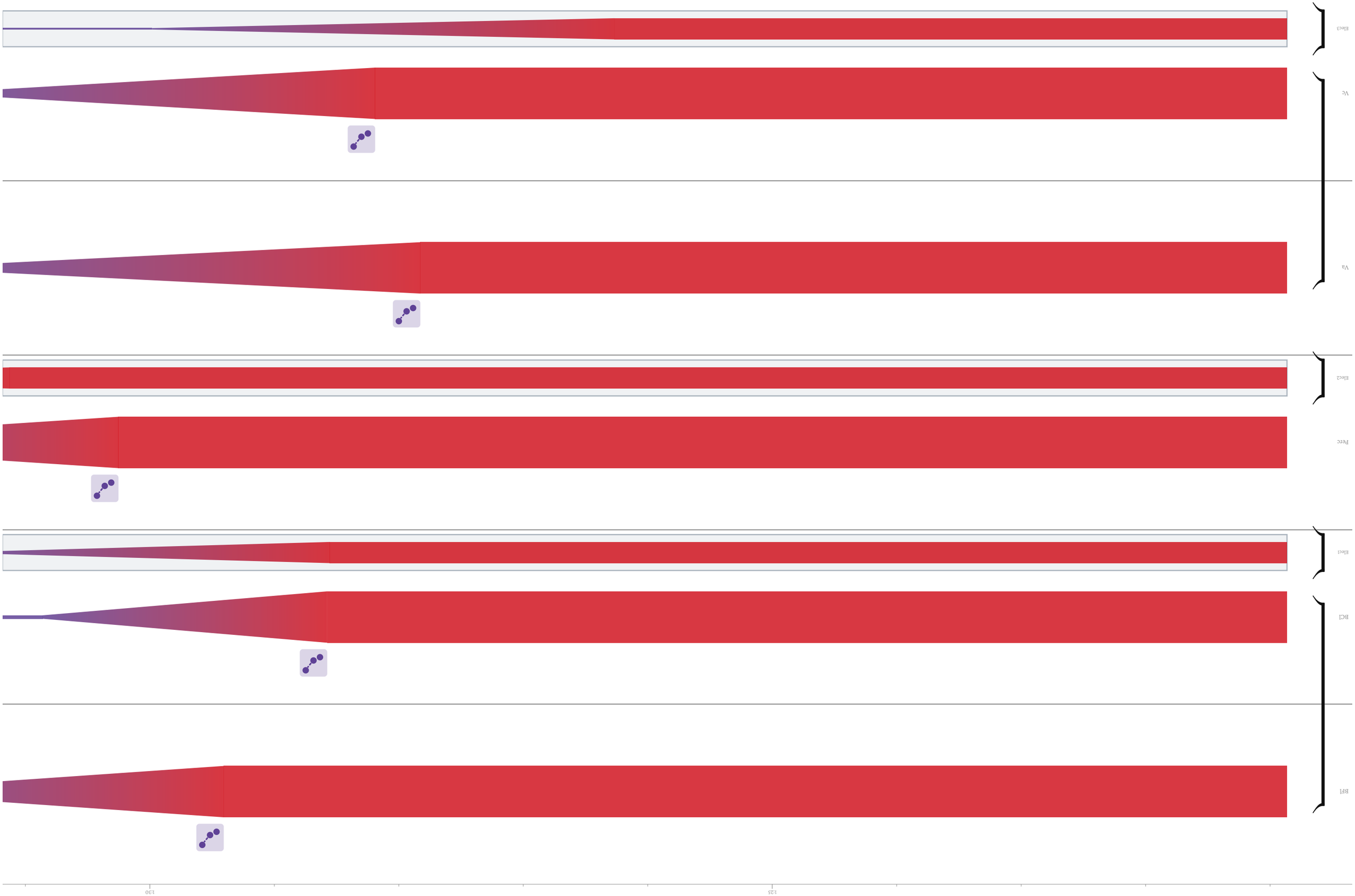
Va

Vc

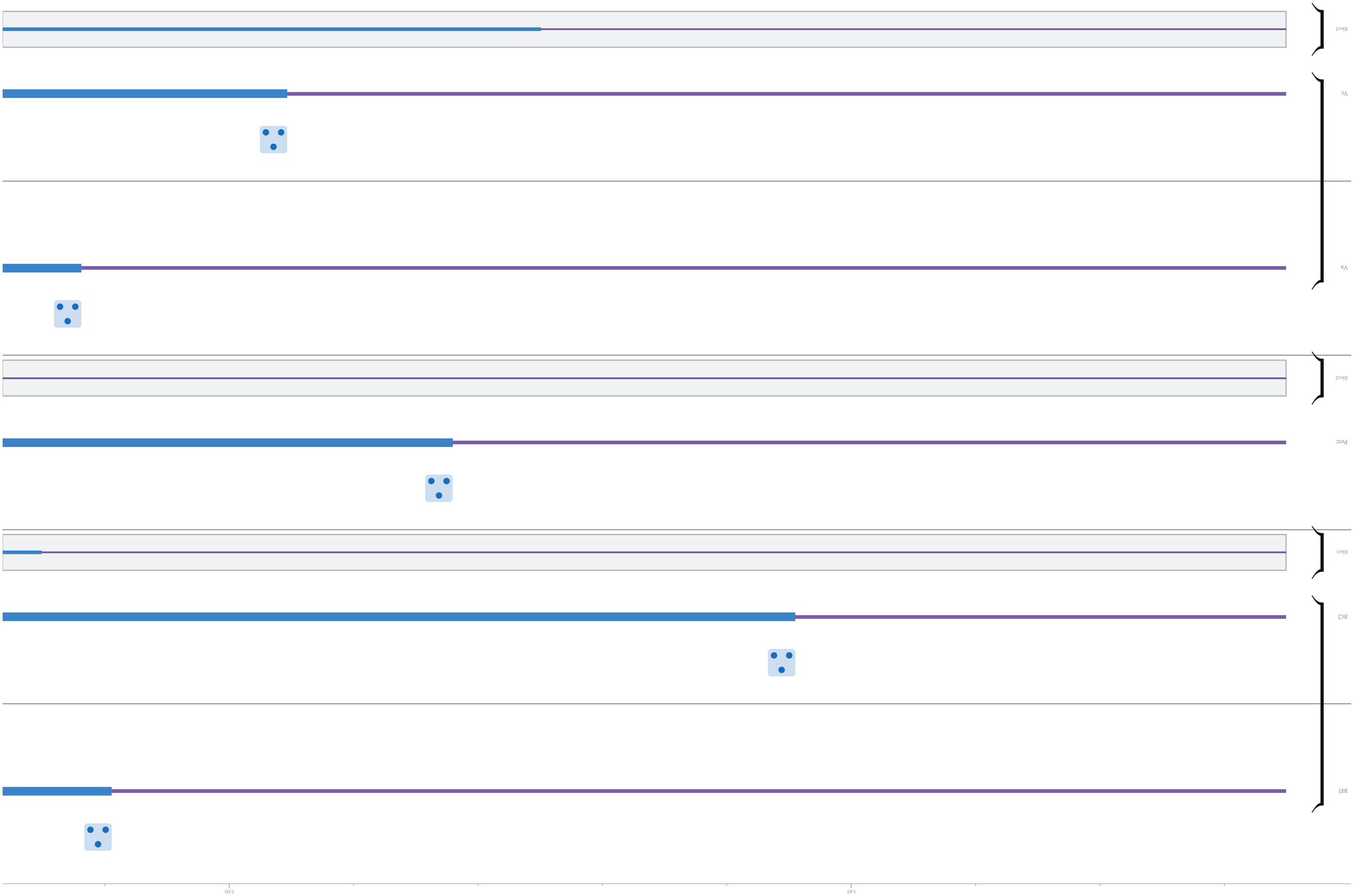
Elec3

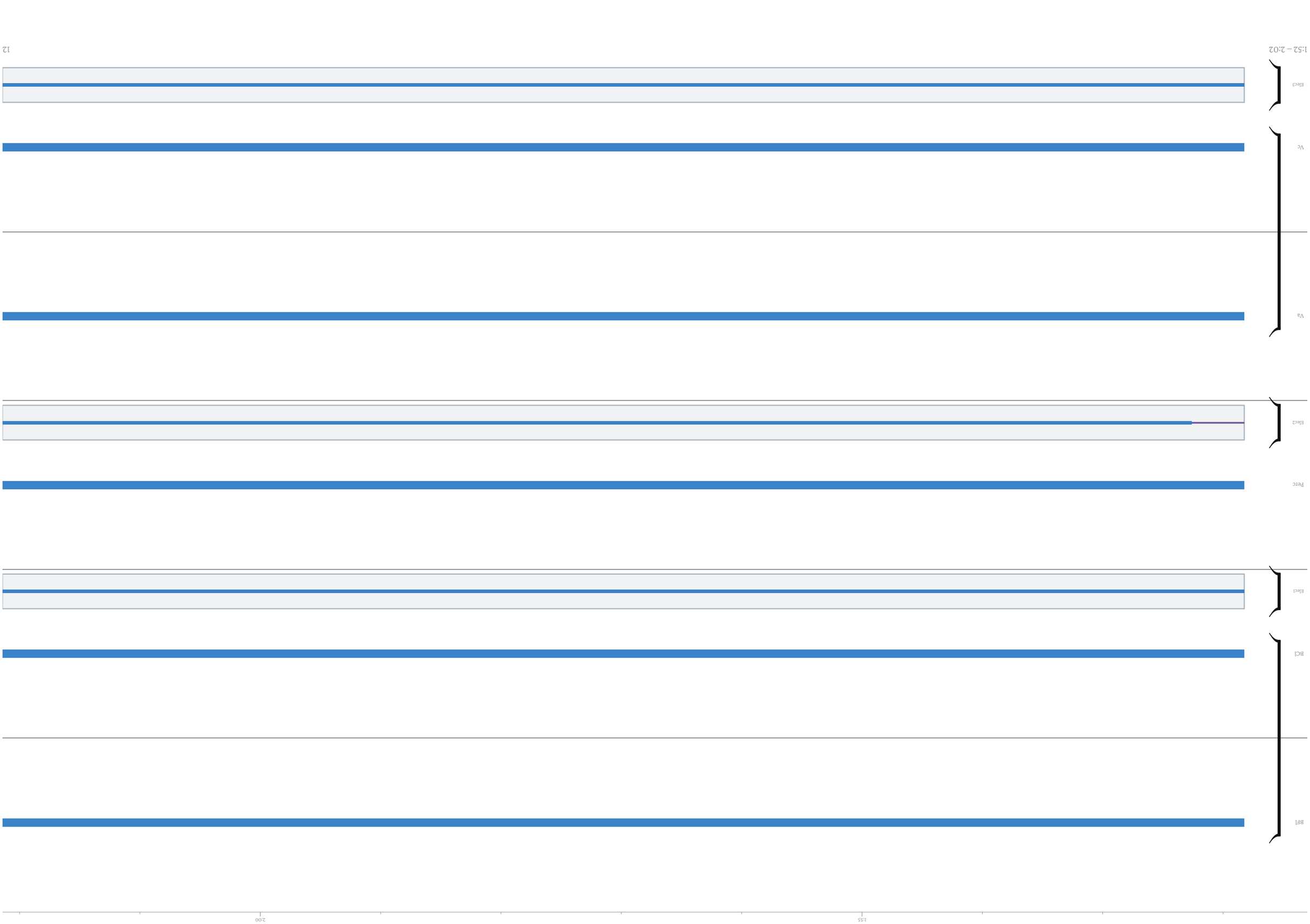


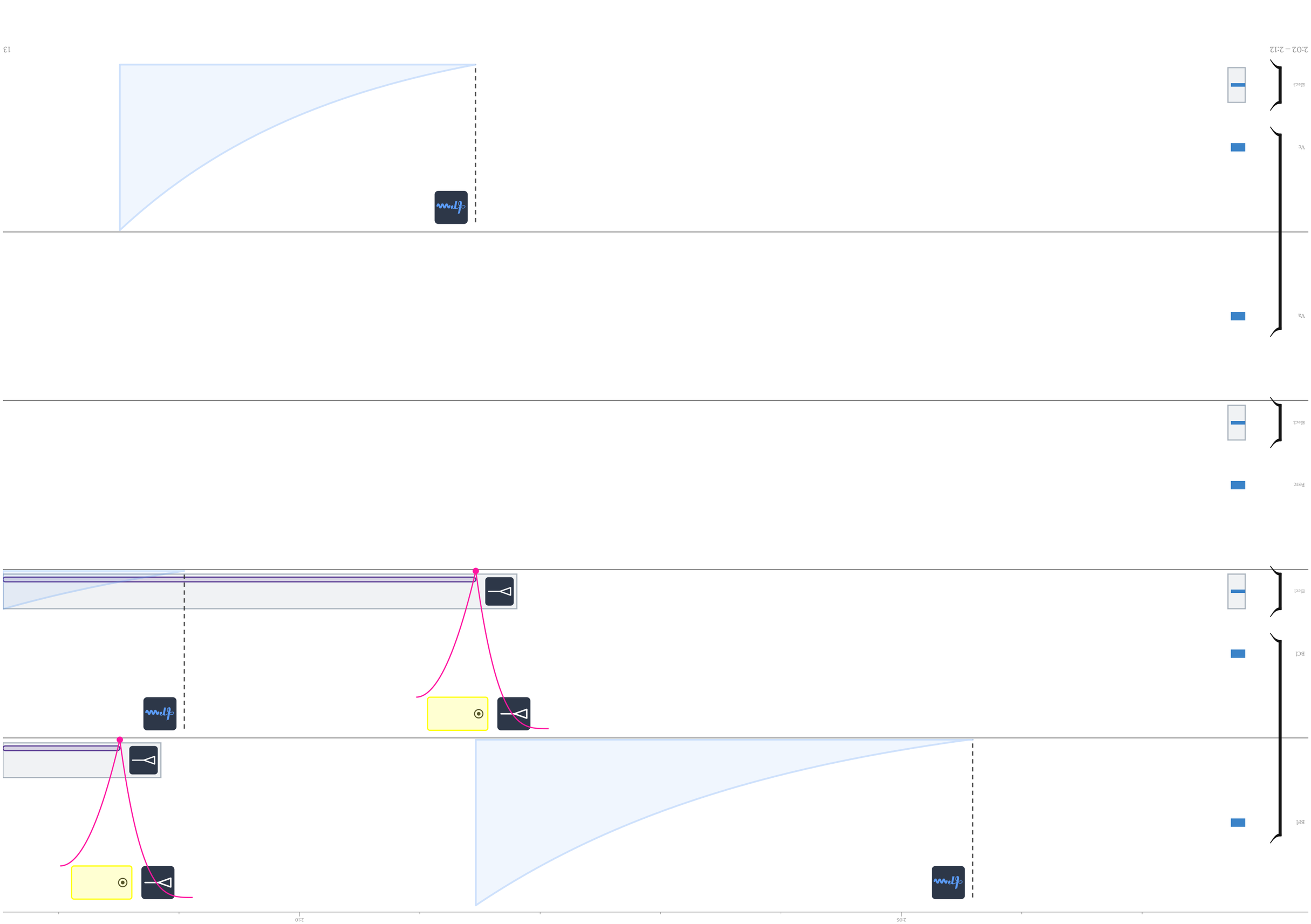
1:11 – 1:21

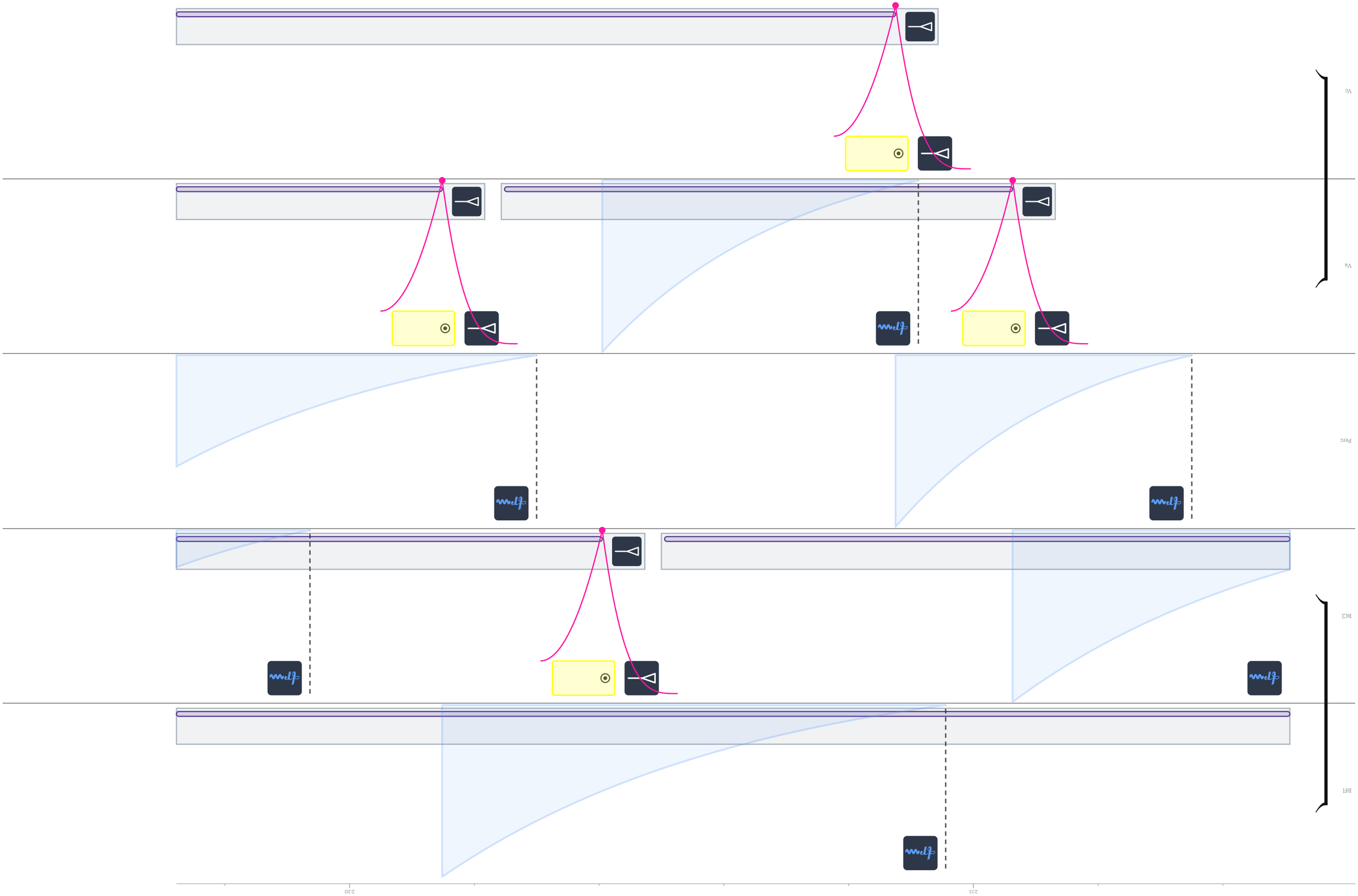

Fetc
Elec2
Va
Vc
Elec3
BF1
BC1
Elec1
1:25
1:30
1:21 – 1:31

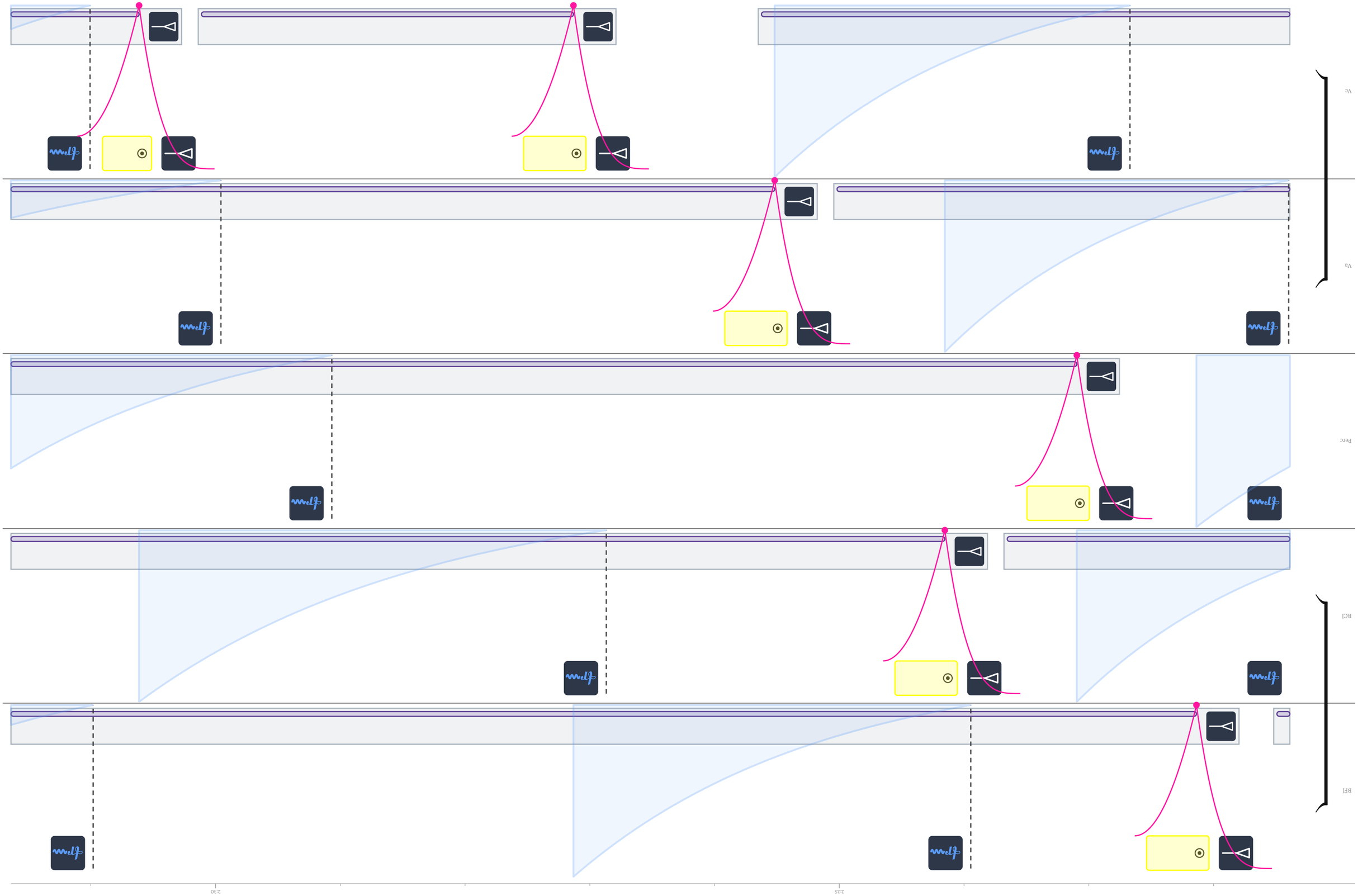

1:31 – 1:42
10
Elec3
Vc
Va
Petc
Elec2
BC1
Elec1
BF1
1:40
1:35

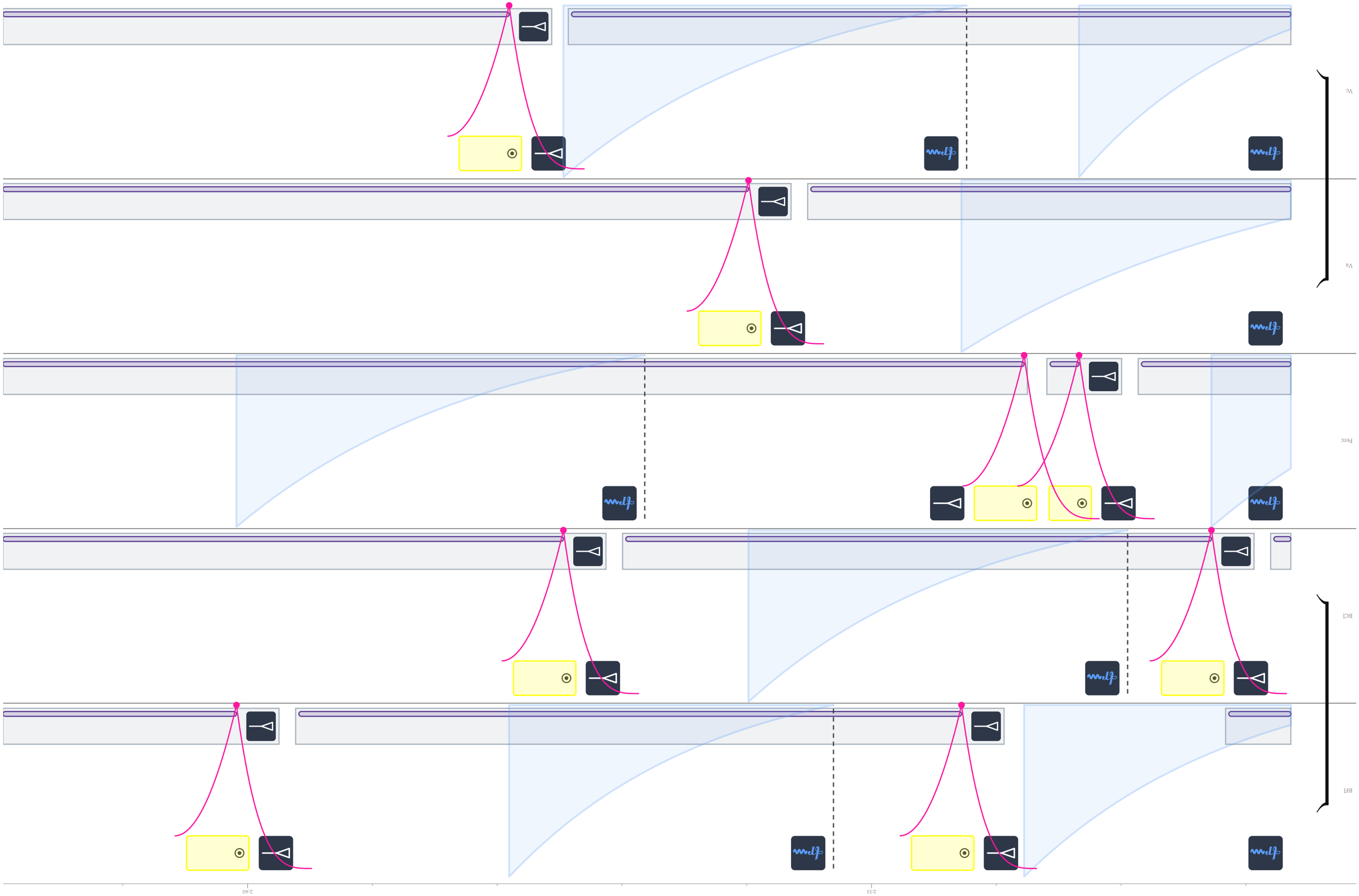

Perc
Elec2
Va
Vc
Elec3
1:42 – 1:52
11
Elec1
BCl
BFl
1:45
1:50

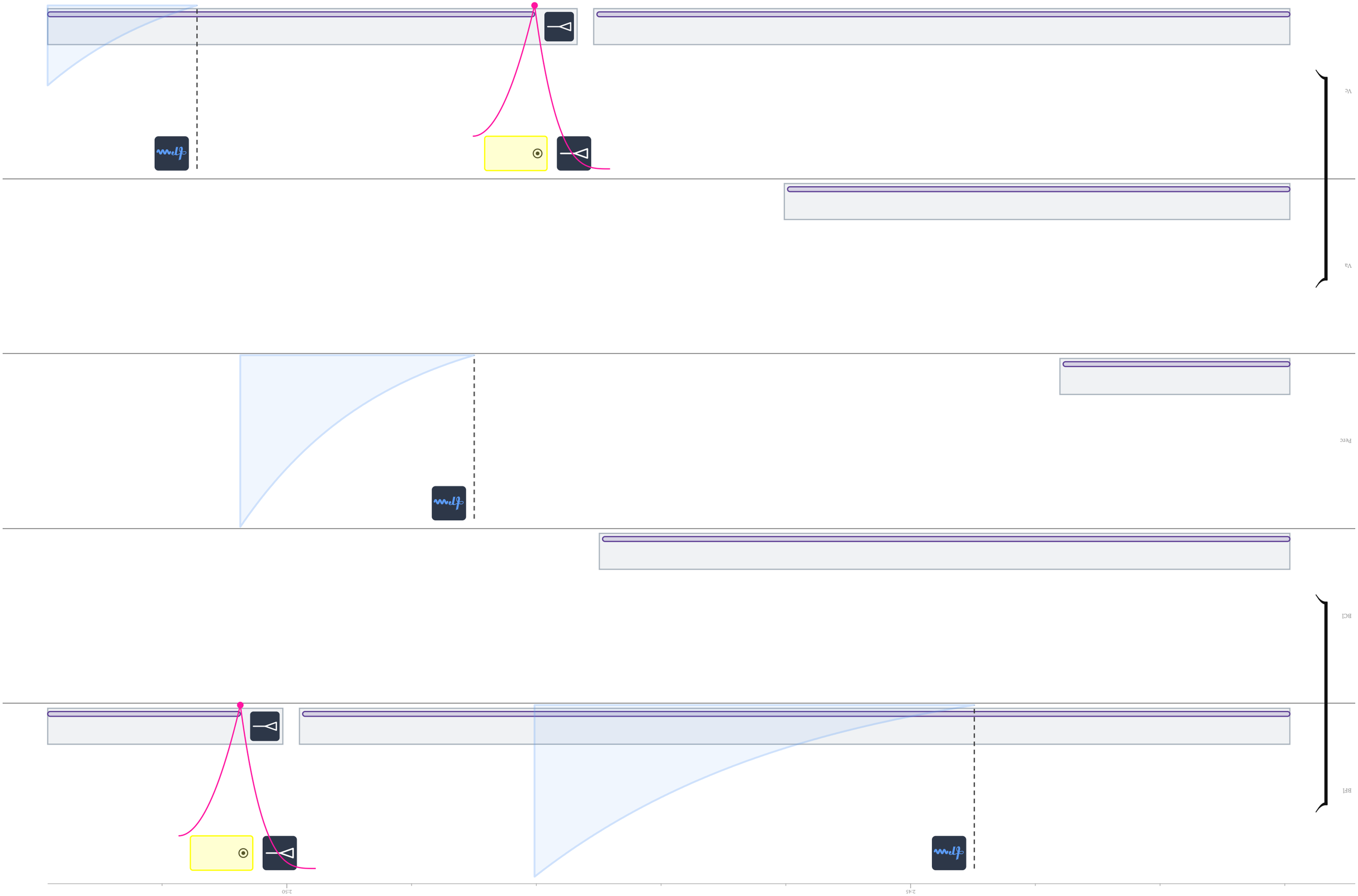

Petc
Elec2
Va
Vc
Elec3
1:52 – 2:02
12
Elec1
BCI
BFI
1:55
2:00


2:02 – 2:12
Elec3
Vc
Va
Petc
Elec2
BCl
BFl
Elec1
2:05
2:10

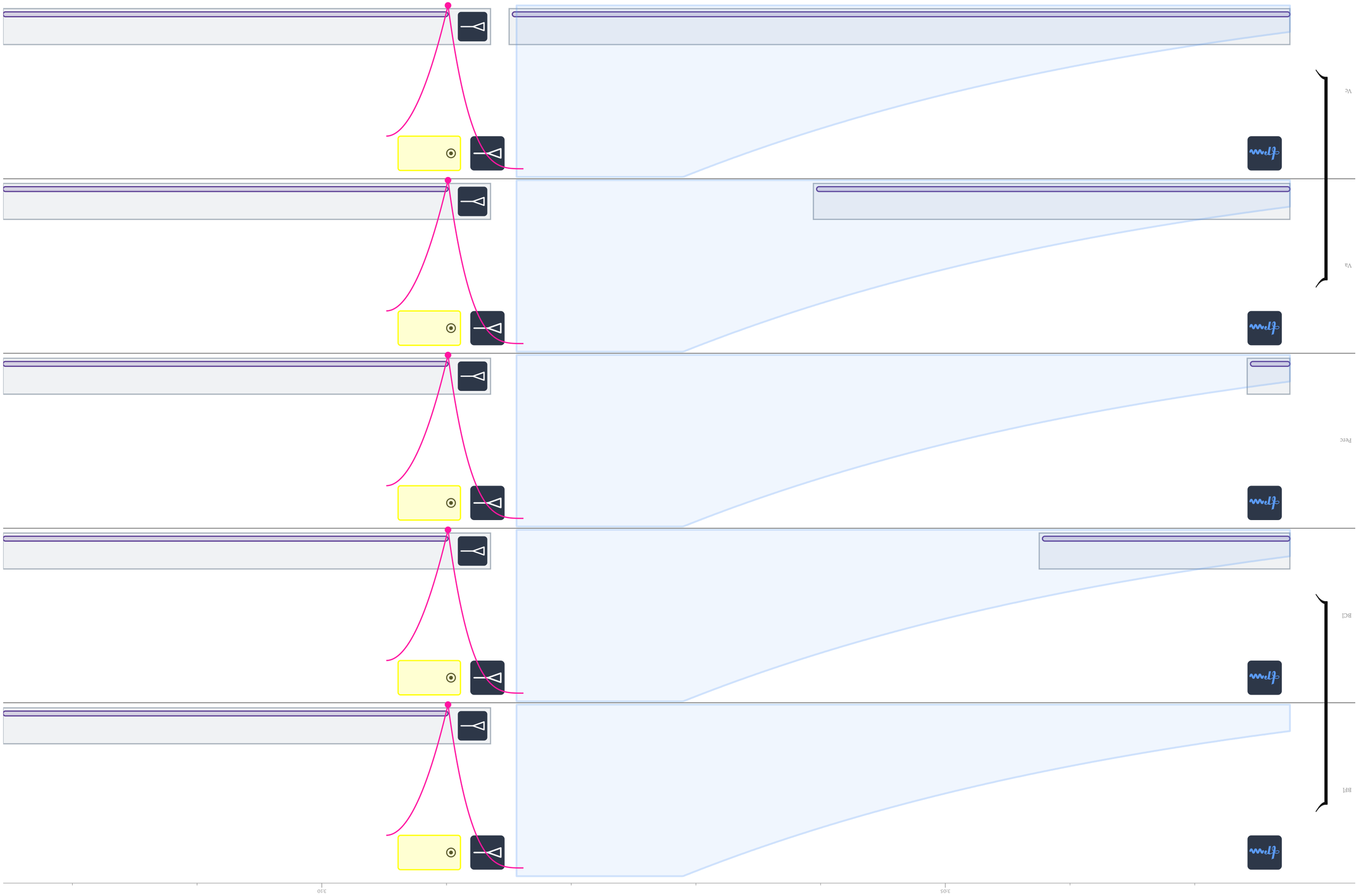

Va
Vc
BCl
BFl
2.20
2.15


Vc
Va
Betc
BCl
BFl
2:30
2:25


Vc
Va
Petc
BCl
BFl
2:40
2:35


Vc
Va
BCl
BFl
2:45
2:50


Vc
Va
Petc
BCl
BFl
2.55
3.00


Vc
Va
Perc
BCl
BFl
3:05
3:10

3:15

3:20

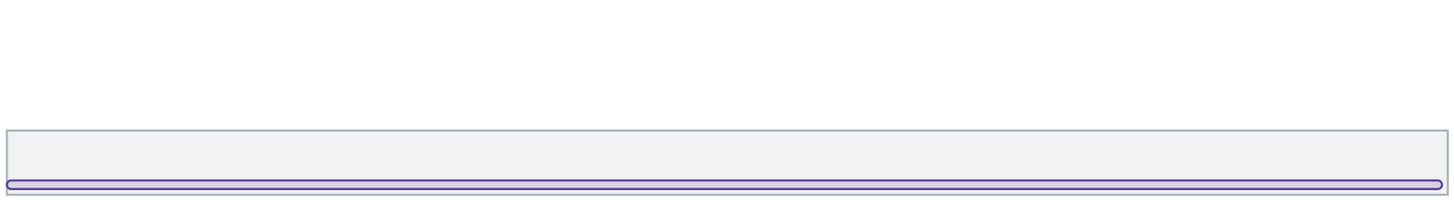
BFl

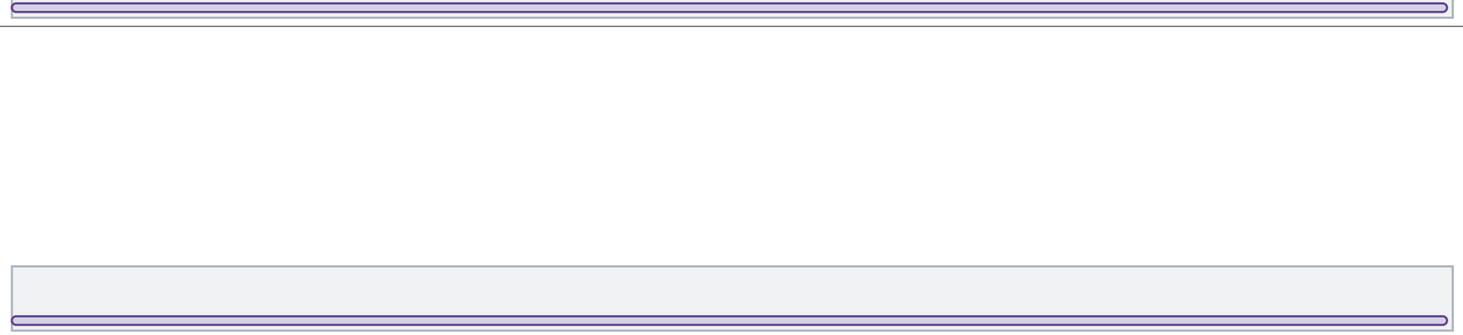
BCl

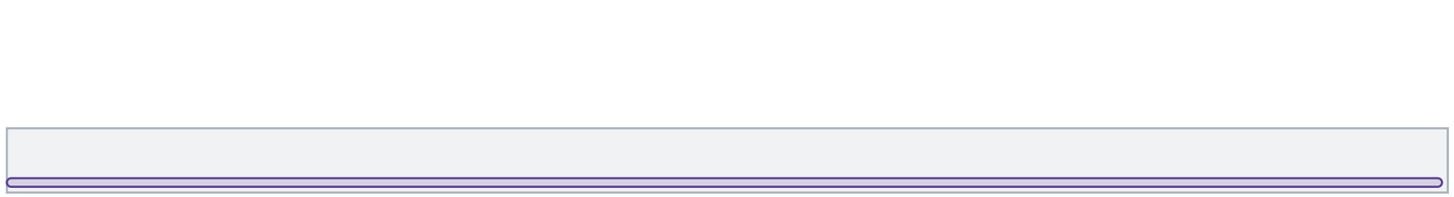
Perc

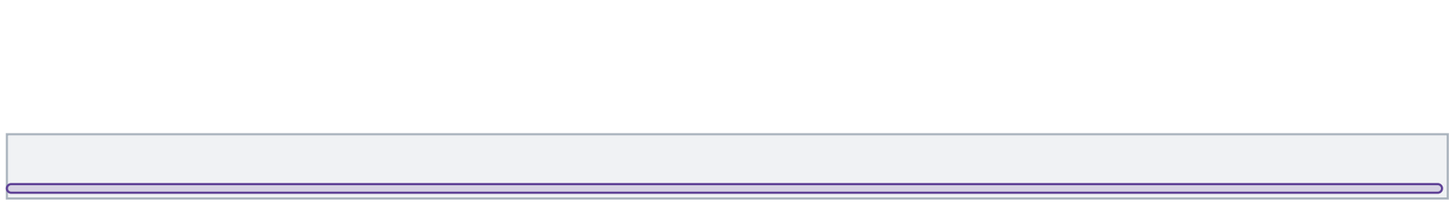
Va

Vc









3:13 – 3:23

3:25

3:30

BFl

BCl

Perc

Va

Vc

3:35

3:40

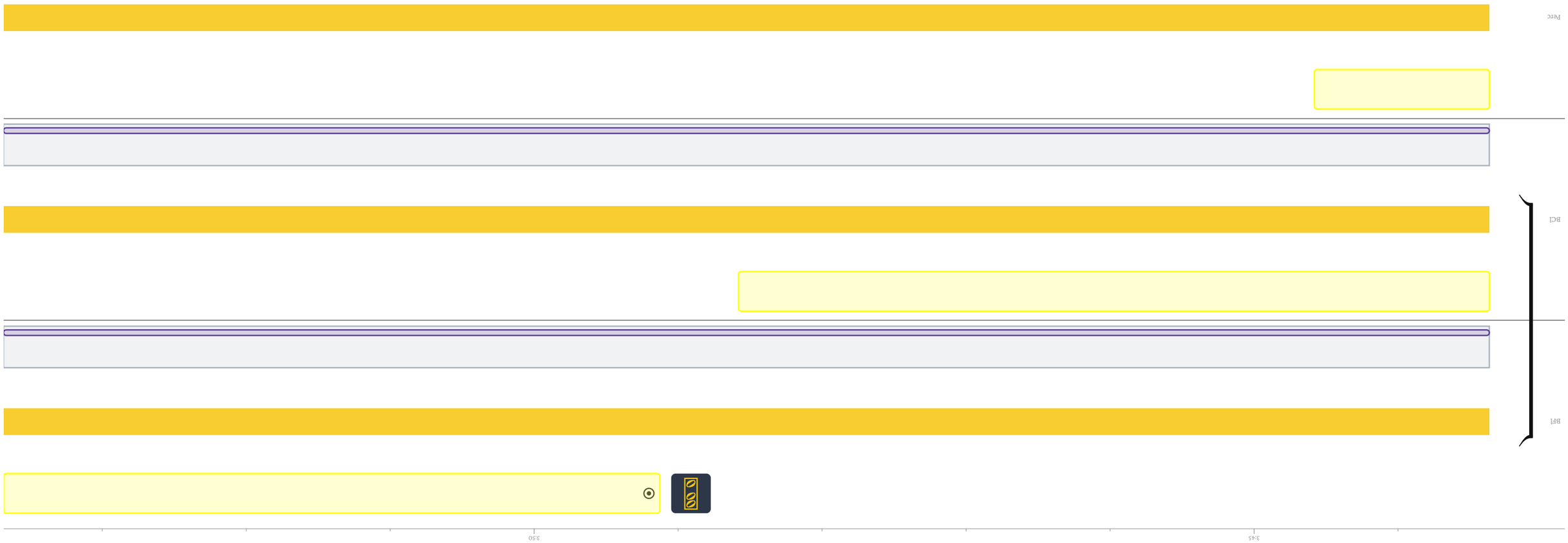
BFl

BCl

Perc

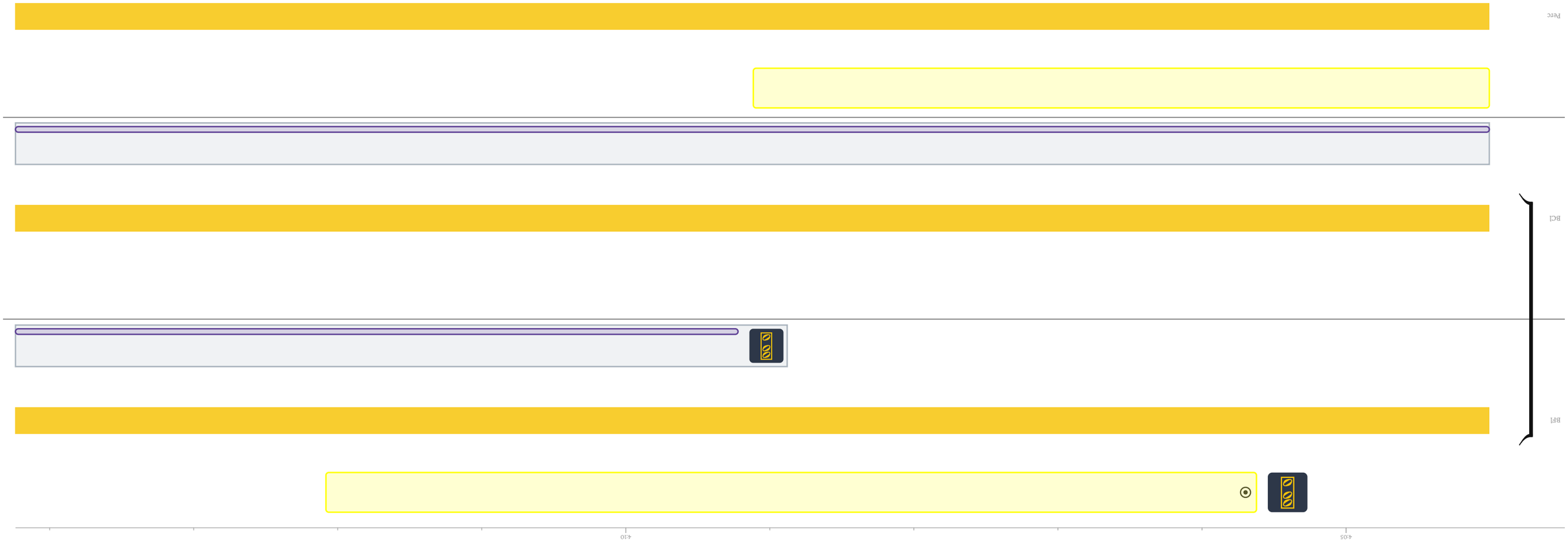
Va

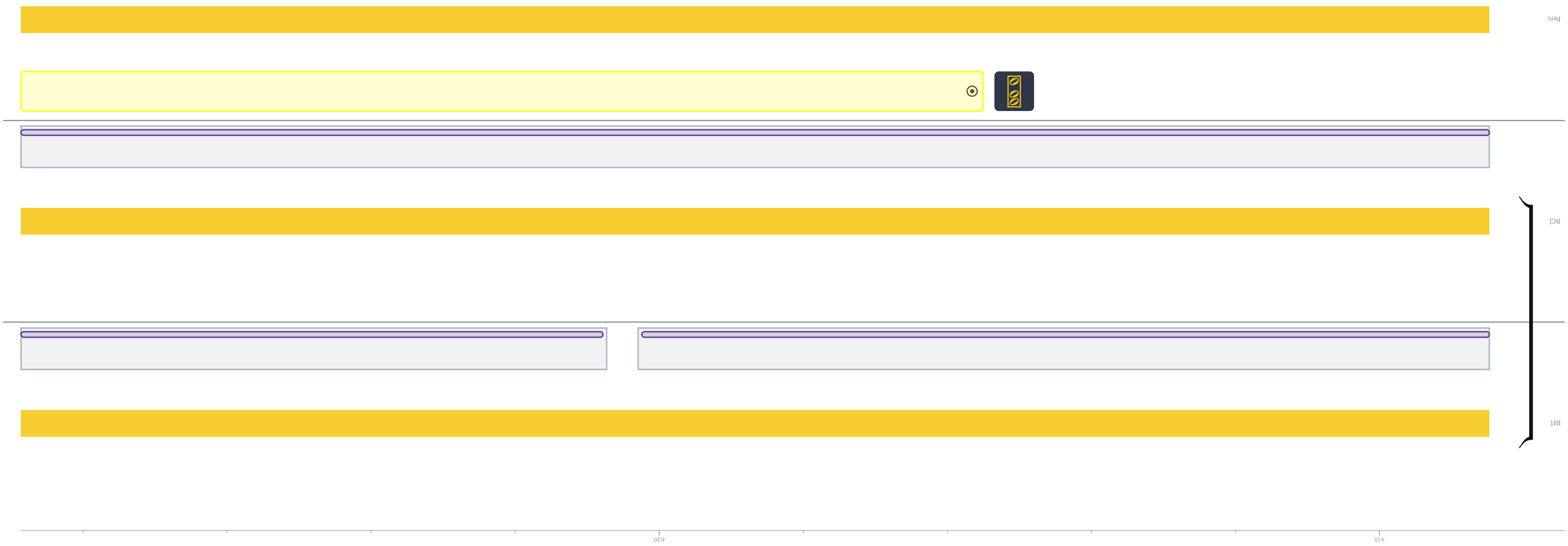
Vc

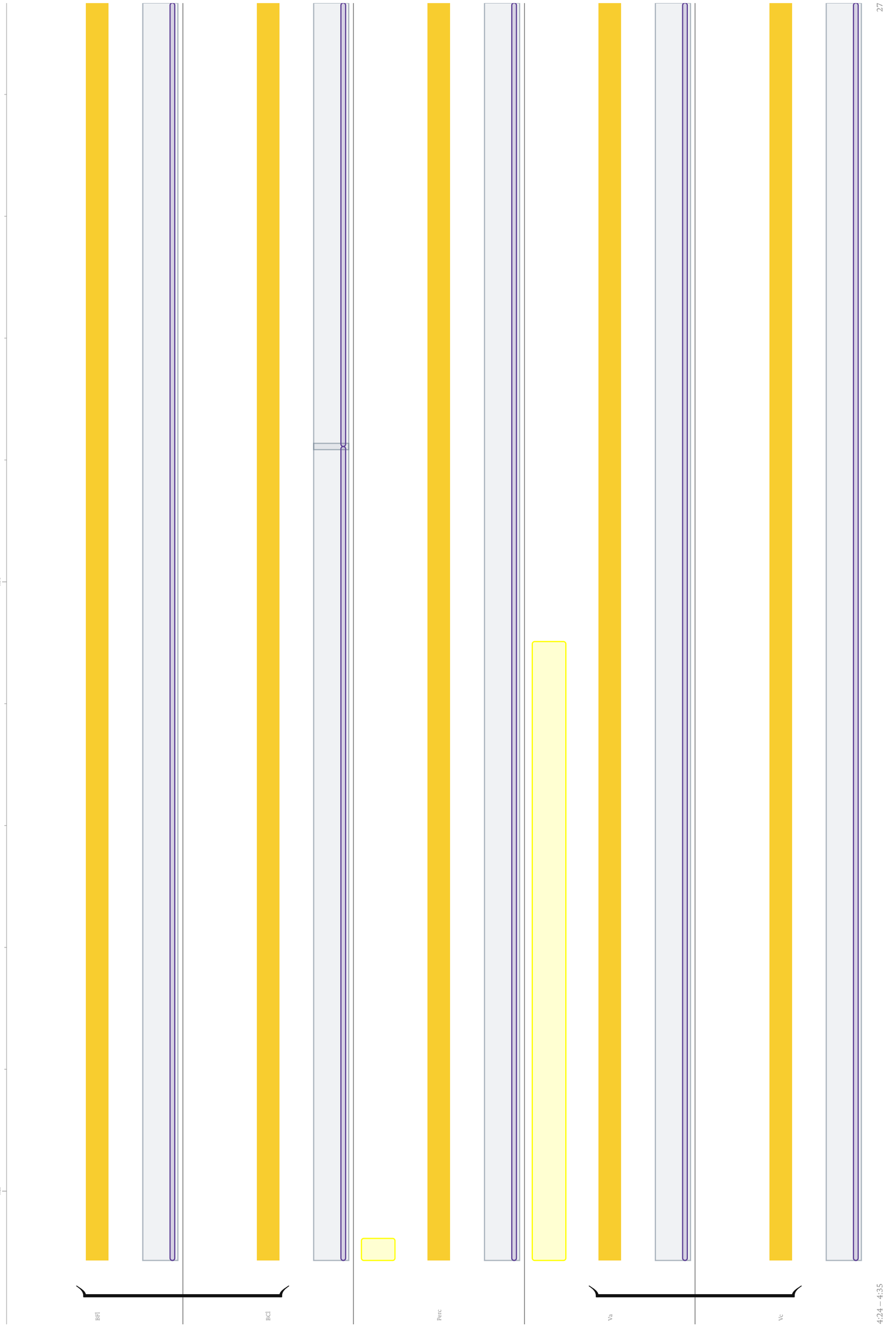


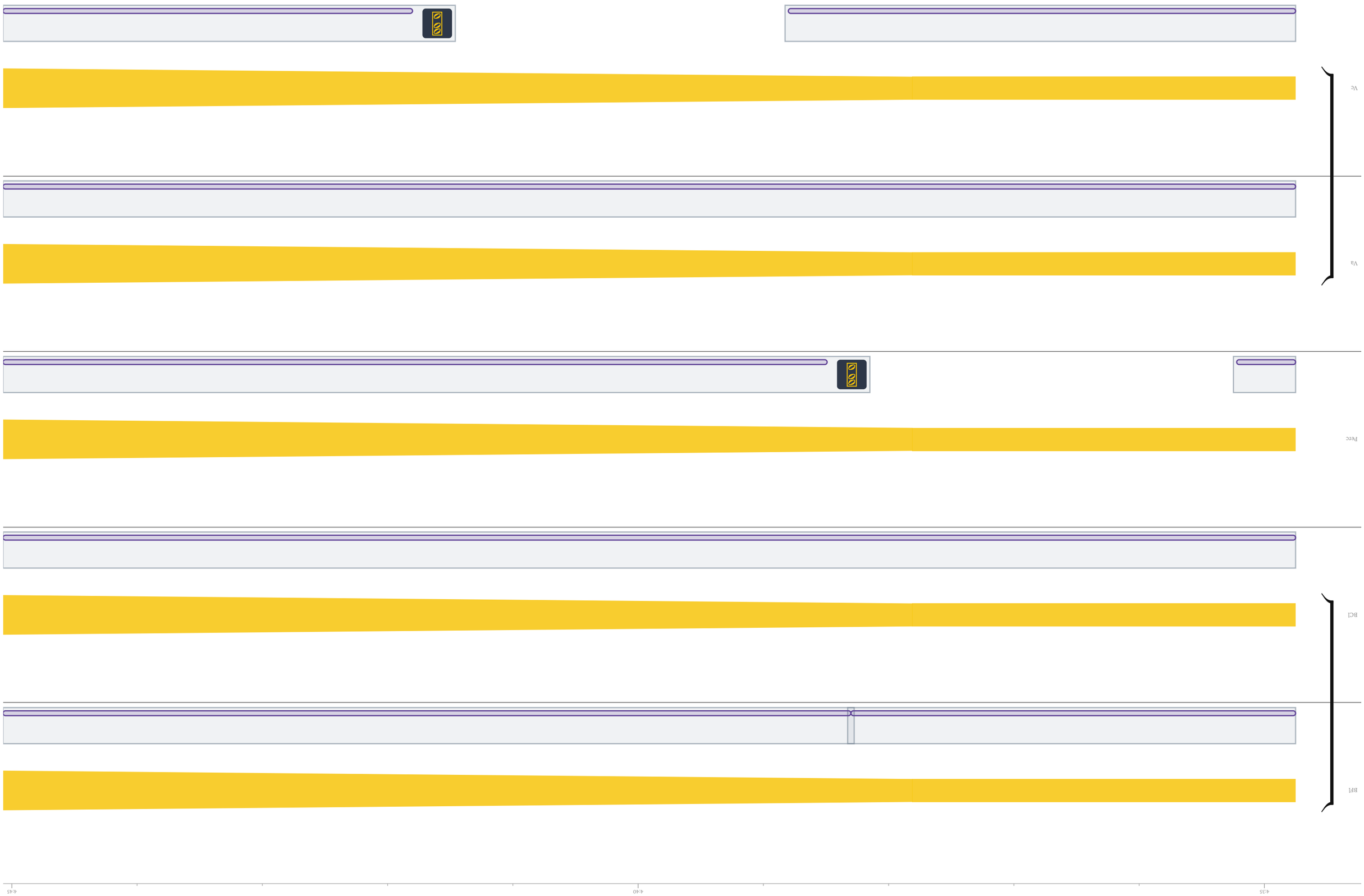
3:43 – 3:54

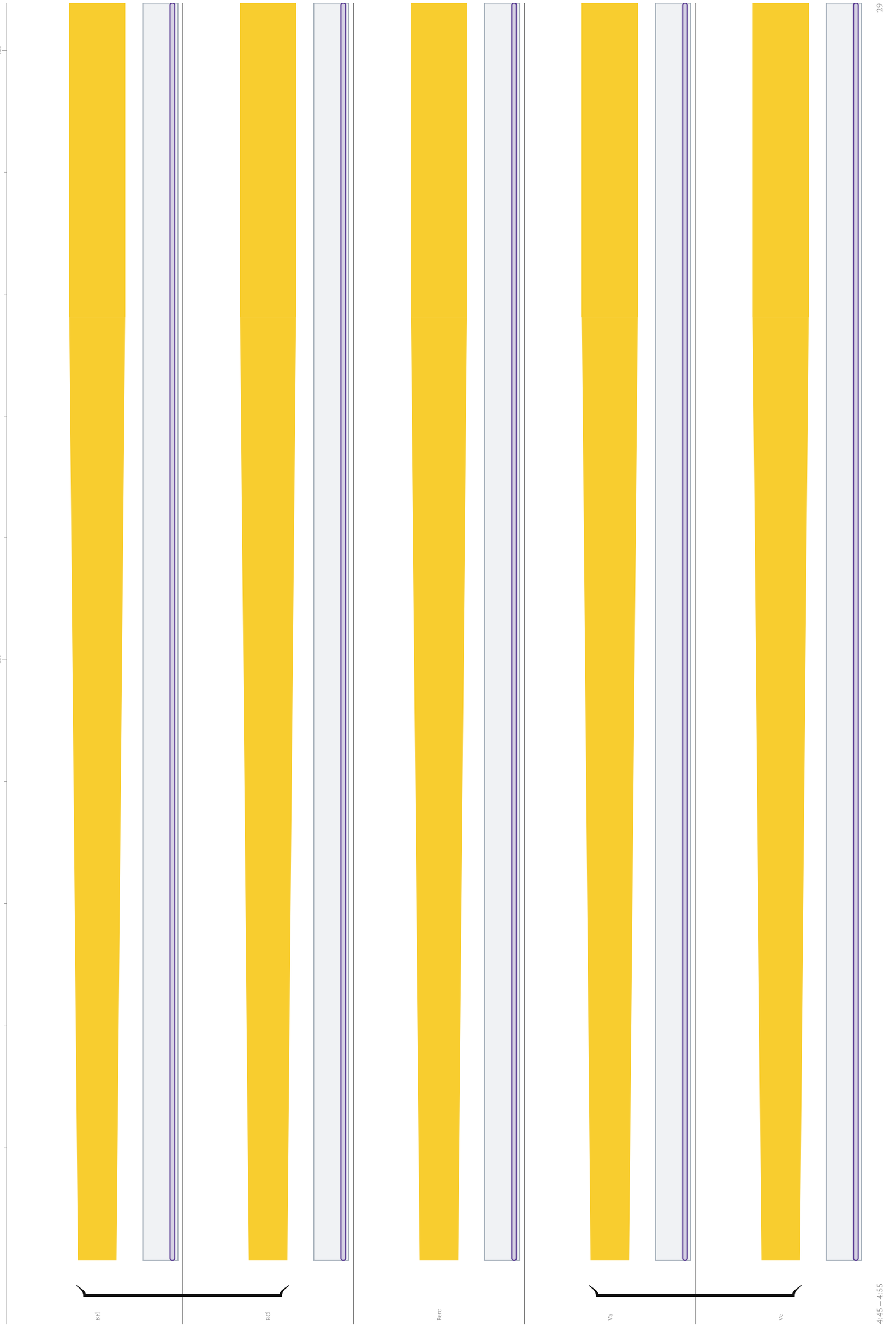

Petc
Va
Vc
BCl
BFl
3:55
4:00


Petc
Va
Vc
BCl
BFl
4:05
4:10


Petc
Va
Vc
BCl
BFl
4:15
4:20


27
4:30
4:25
BFl
BCl
Perc
Va
Vc
4:24 – 4:35


Fetc
Va
Vc
BF1
BC1
4:35
4:40
4:45



4:45 – 4:55

5:00

5:05

BFl

BCl

Perc

Va

Vc

5:10

5:15

BFl

BCl

Perc

Va

Vc

BFl

BCl

Perc

Va

Vc

5:30

5:35

BFl

8va

BCl

Perc

Va

Vc

5:40

5:45

BFl

BCl

Perc

Va

Vc

5:50

5:55

BFl

BCl

Perc

Va

Vc

6:00

6:05

BFl

BCl

Perc

Va

Vc

BFl

BCl

Perc

Va

Vc

6:20

6:25

BFl

BCl

Perc

Va

Vc

6:30

6:35

BFl

BCl

Perc

Va

Vc

6:40

6:45

BFl

BCl

Perc

Va

Vc

6:50

6:55

BFl

BCl

Perc

Va

Vc

7:00

7:05

BFl

BCl

Perc

Va

Vc

7:10

7:15

BFl

BCl

Perc

Va

Vc

7:20

7:25

BFl

BCl

Perc

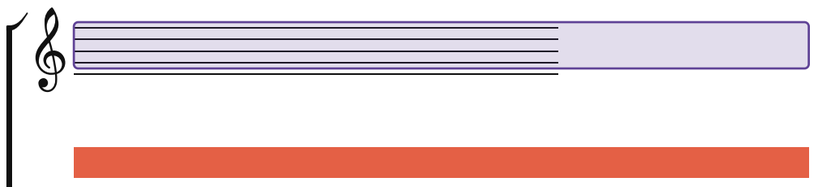
Va

Vc

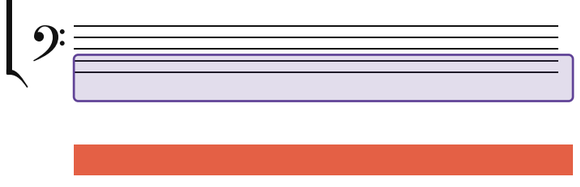
7:30

7:35

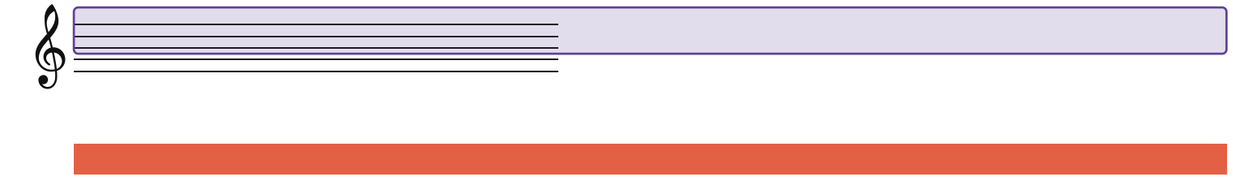
BFl



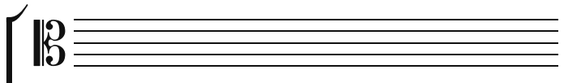
BCl



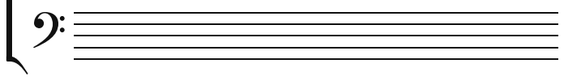
Perc

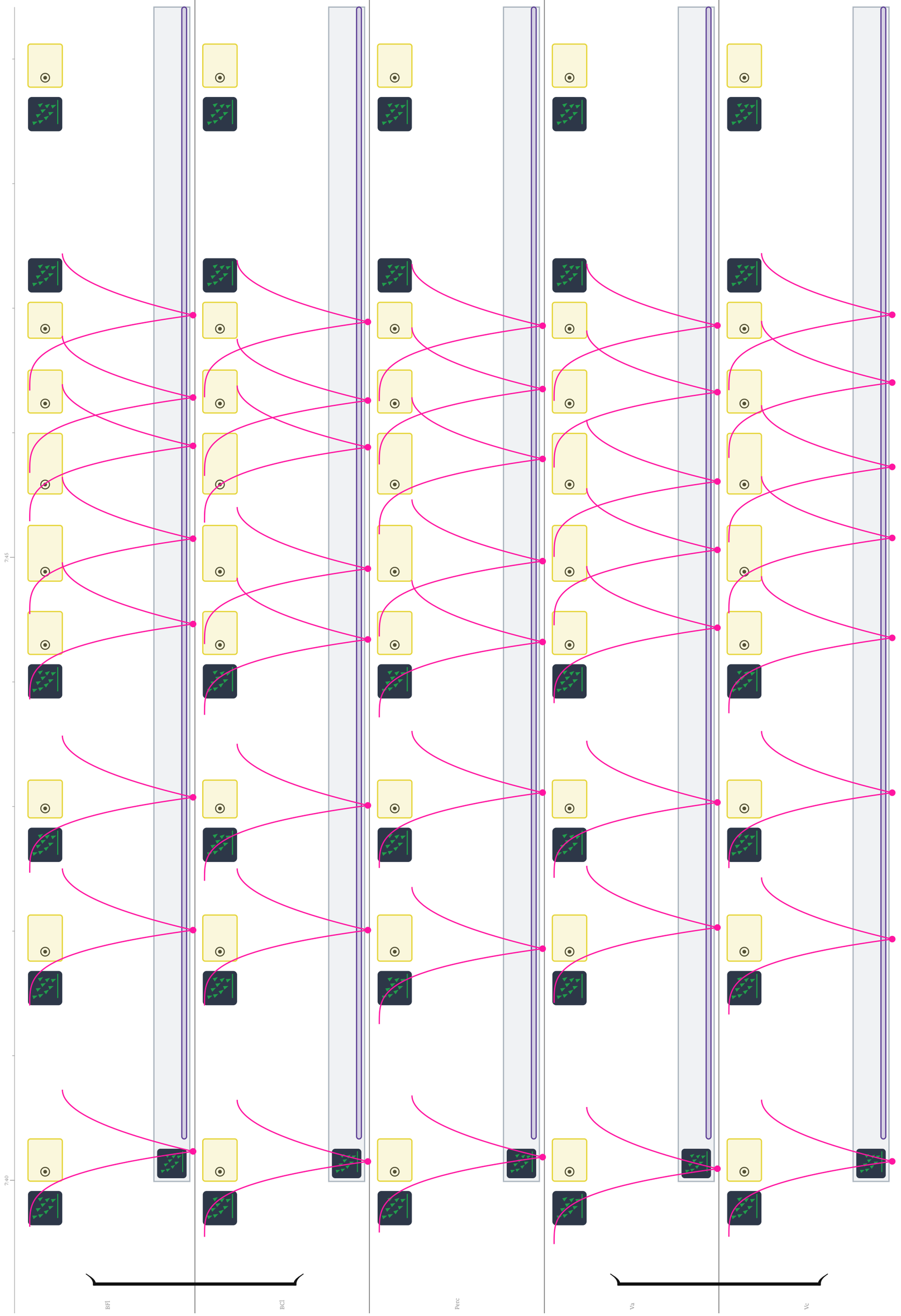


Va



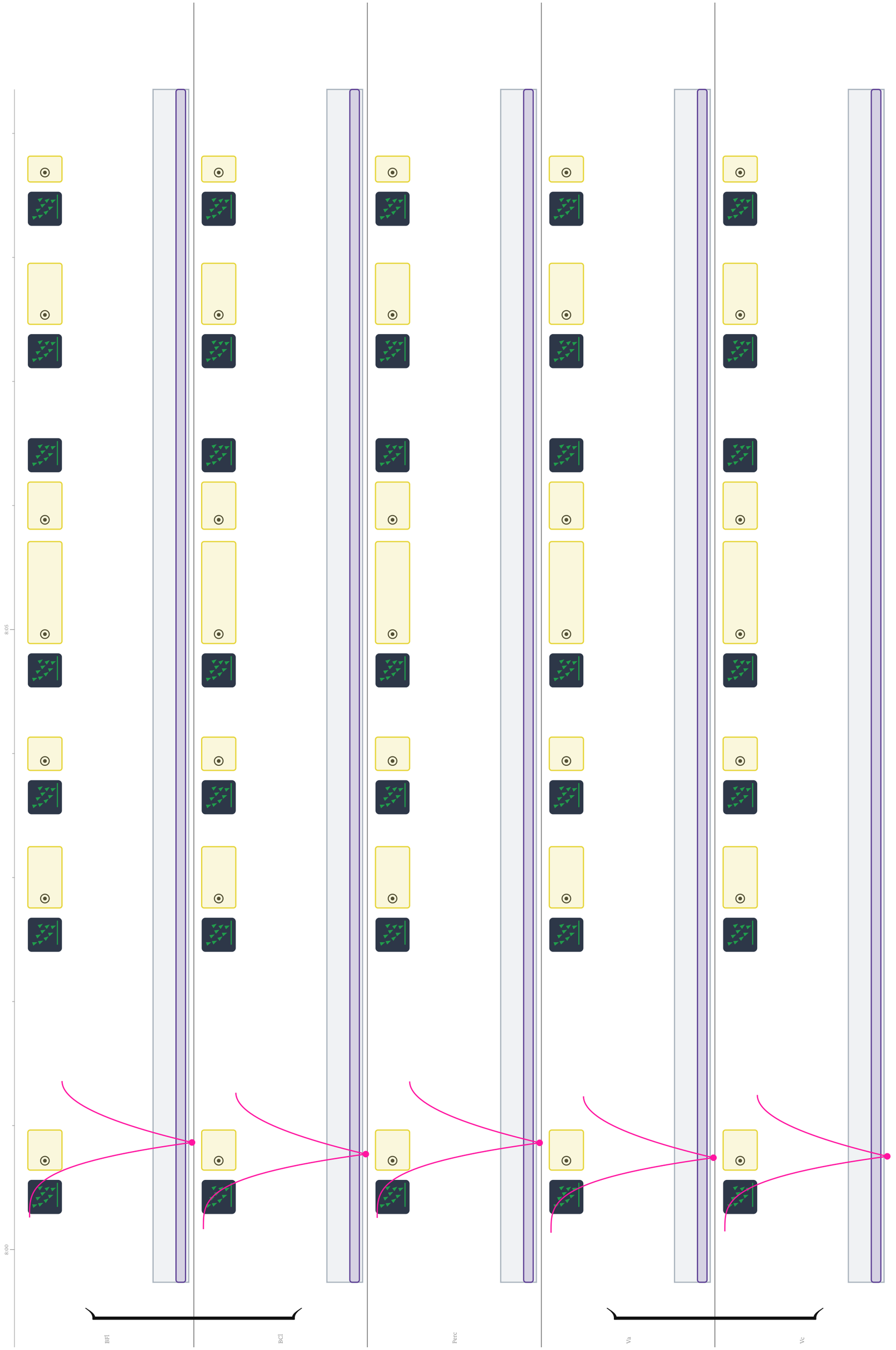
Vc



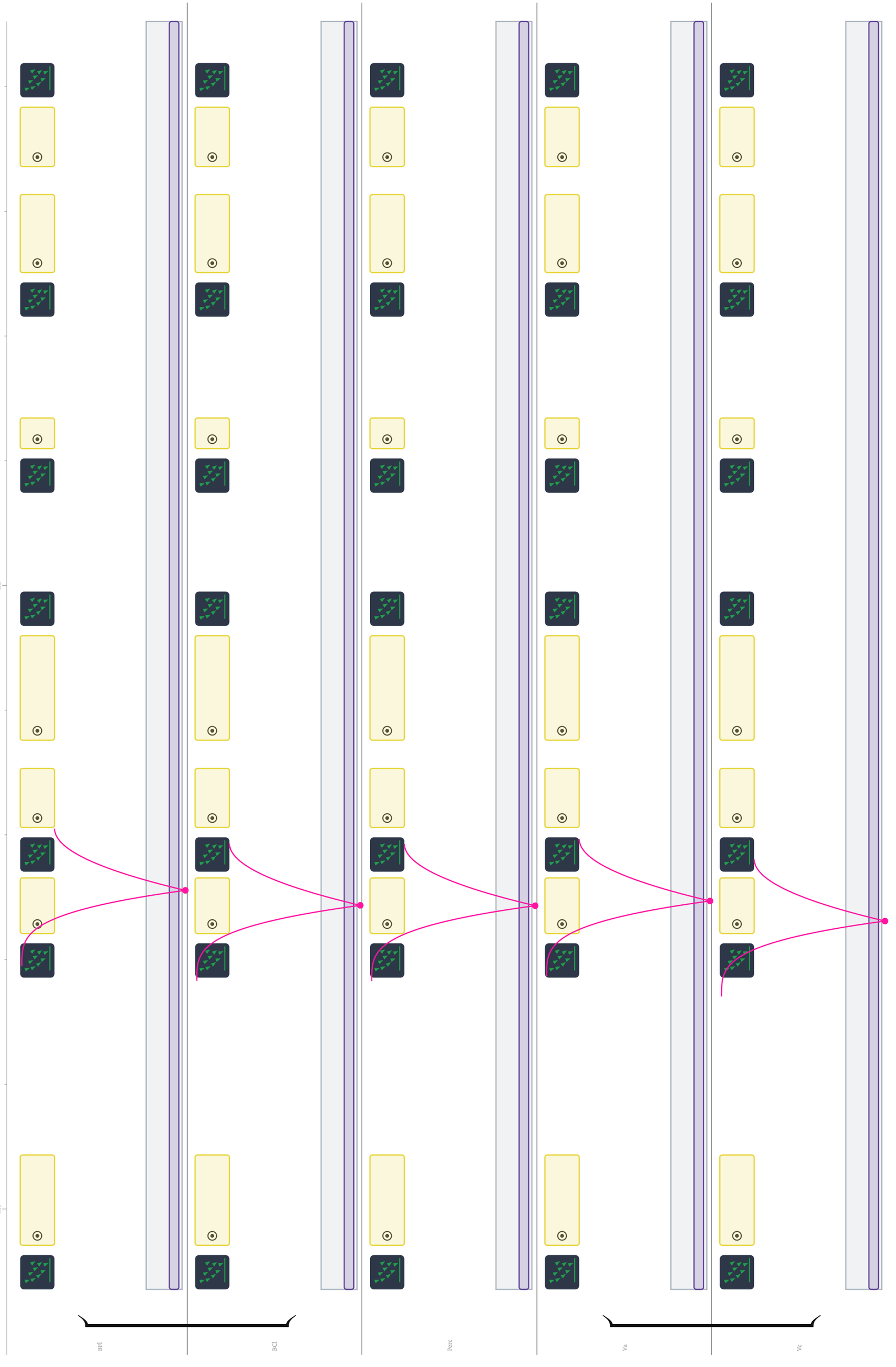


7:39 – 7:49

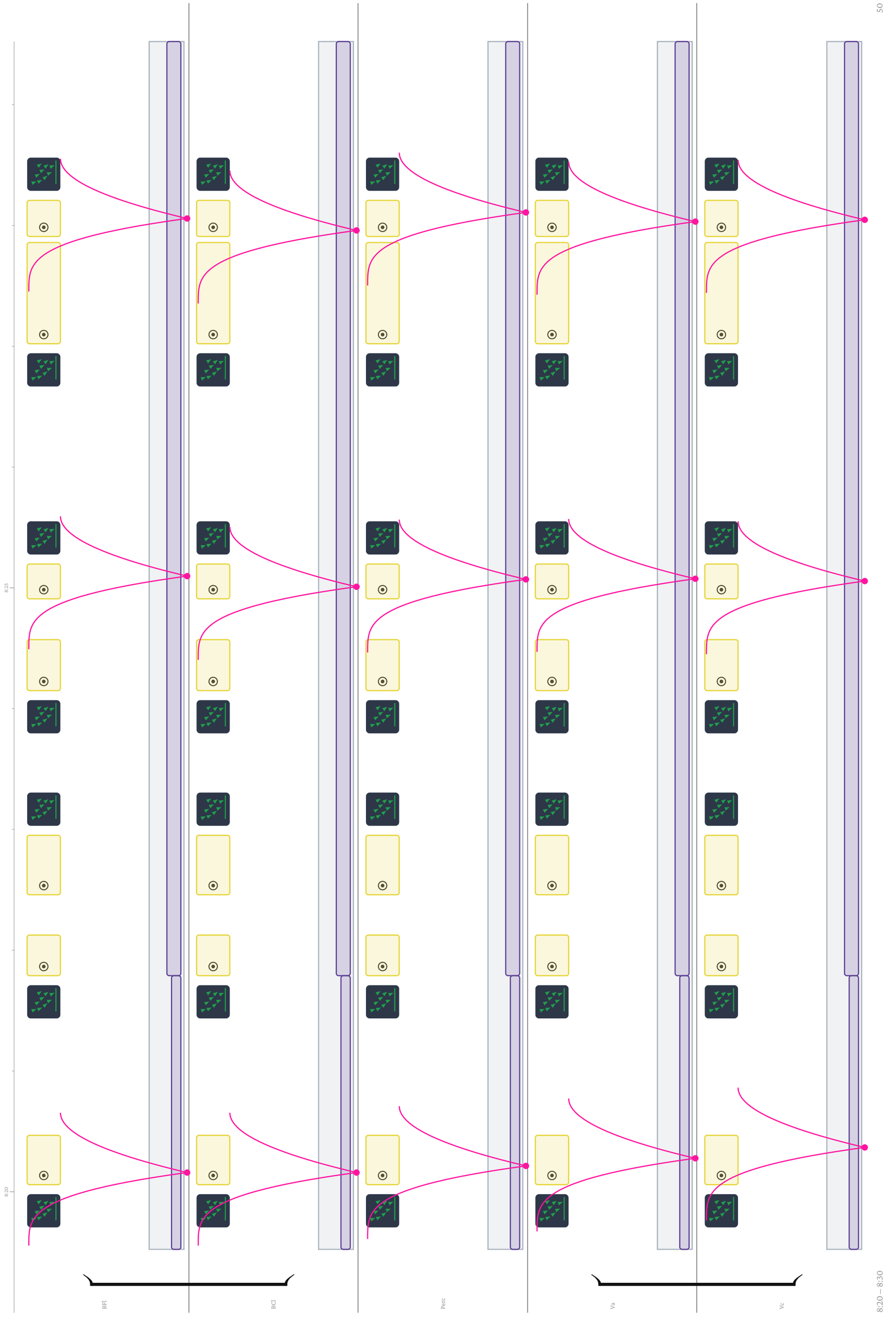

Vc
Va
BCI
BFI
Petc
7:50
7:55

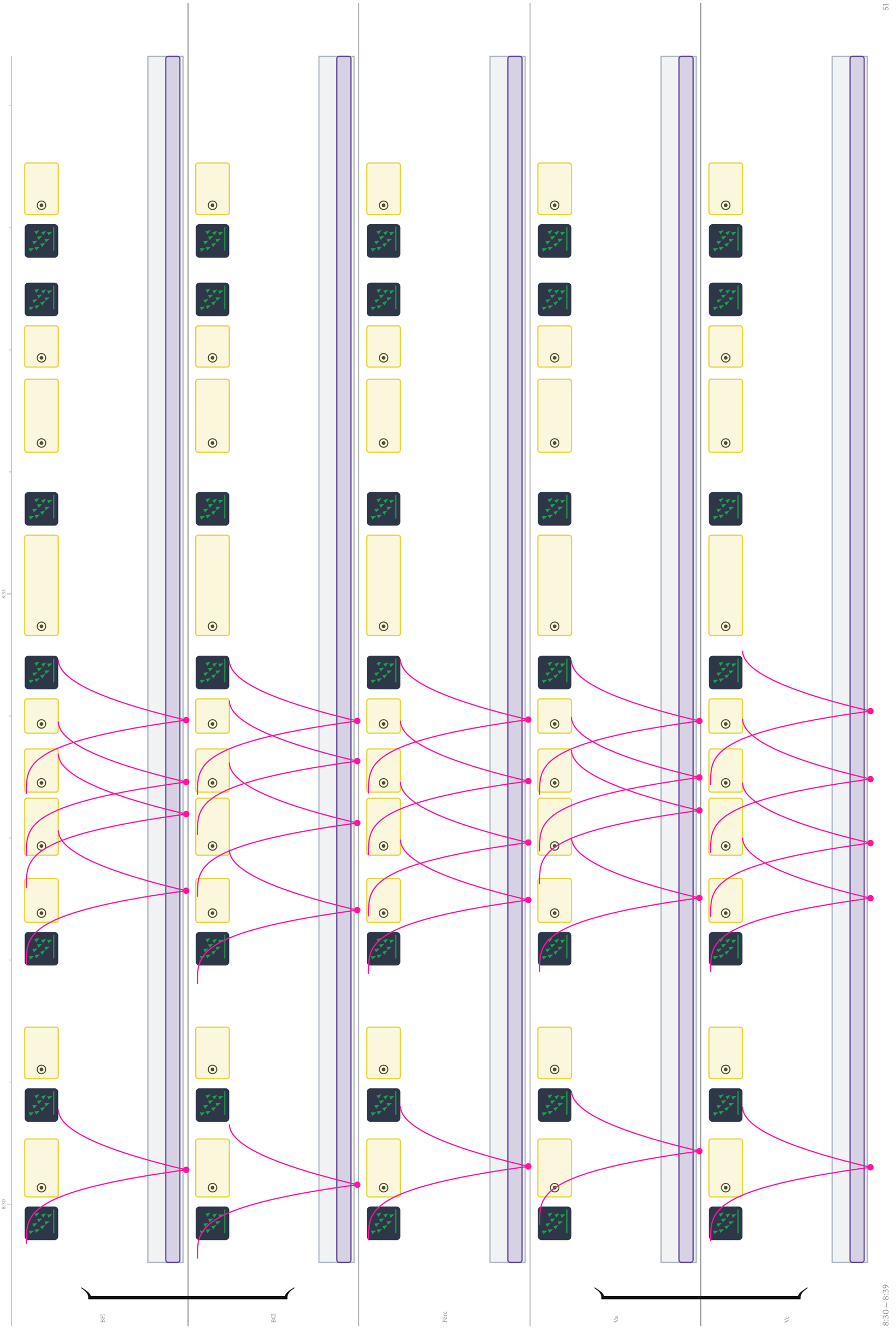


8:00 – 8:09



8:09 – 8:20


8:20
8:25
50
BF1
BC1
Perc
Va
Vc
8:20 – 8:30


8:35
8:30
BFl
BCl
Perc
Va
Vc
8:30 – 8:39

8:40

8:45

BFl

BCl

Perc

Va

Vc

8:39 – 8:49

8:55

8:50

BFl

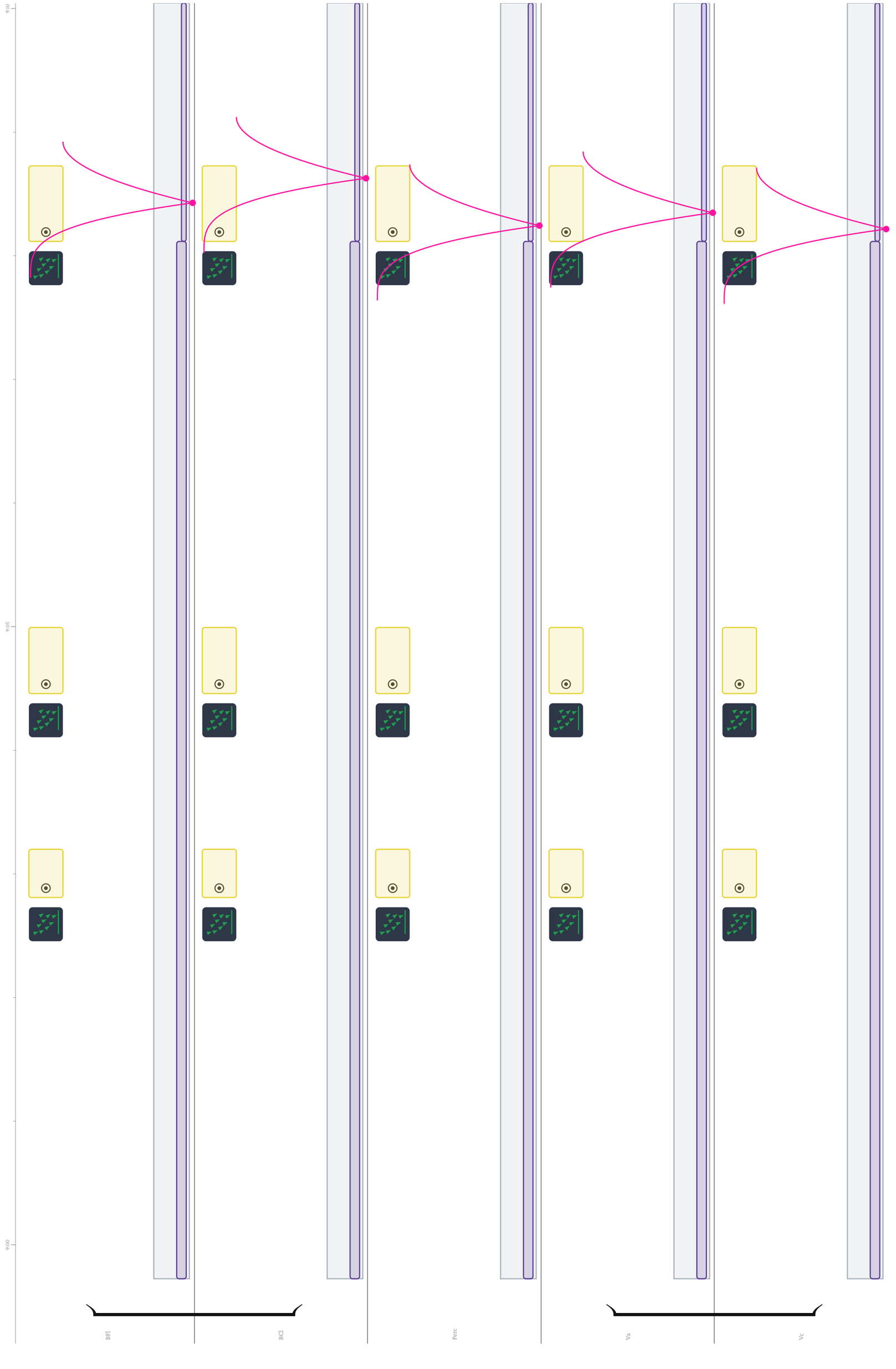
BCl

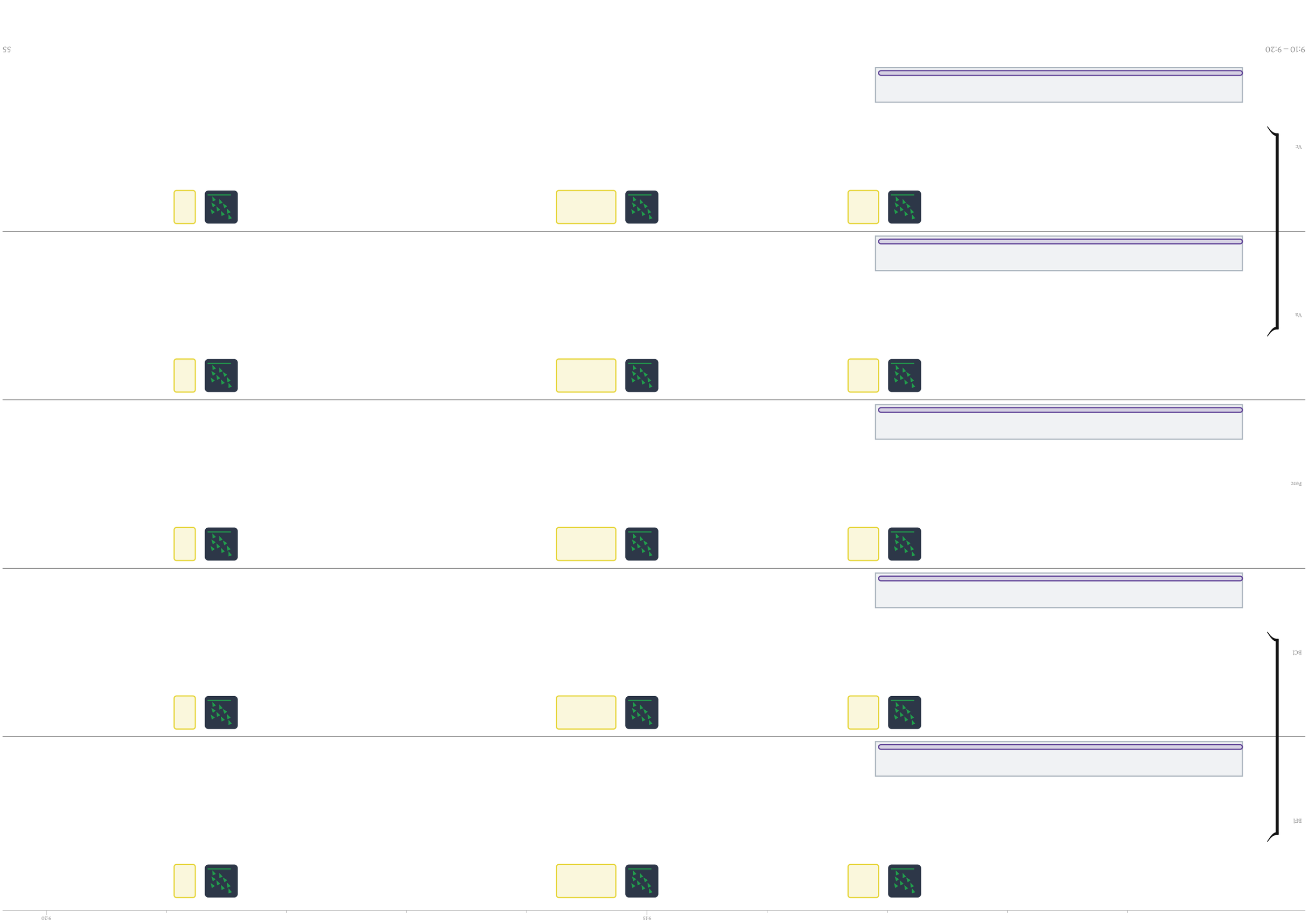
Perc

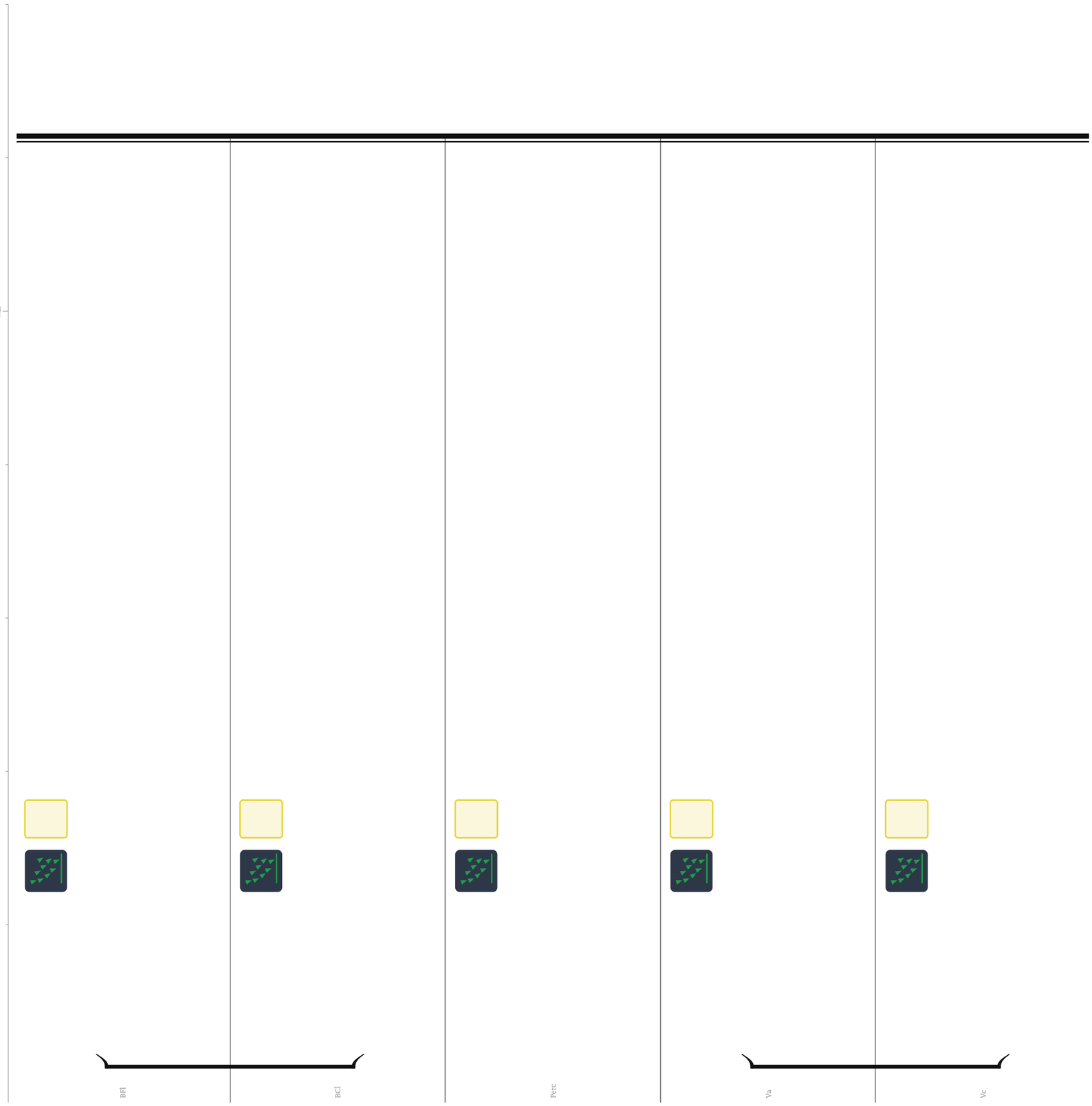
Va

Vc

8:49 – 9:00


9:10
9:05
9:00
BFl
BCl
Perc
Va
Vc
9:00 – 9:10


Fetc
Va
Vc
BCl
BFl
9.10 – 9.20
9.15
9.20


9:25
Bfl
Bcl
Perc
Va
Vc
9:20 — 9:27